Robust Aggregation of Expert Probability Forecasts via Wasserstein Barycenters

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Abstract

We study the aggregation of expert probability forecasts over a finite outcome space. The standard approach, the arithmetic mean (linear opinion pool), is sensitive to extreme opinions: a minority assigning substantial mass to tail outcomes can distort the aggregate away from the majority view. We propose the Wasserstein barycenter as a robust alternative. Under the indicator ground cost the barycenter reduces to an ℓ^1 aggregation problem; for ordered outcome spaces it admits a closed form—the distribution whose CDF is the component-wise median of the individual CDFs—computable without a solver. We validate the framework across three datasets. On 94 binary questions from the Good Judgment Project ($\sim$900 forecasters each), the barycenter reduces mean Brier score by 12.5% relative to the arithmetic mean ($p < 0.001$). On 76 quarters of U.S. Survey of Professional Forecasters core CPI histograms (2007–2025), the barycenter achieves a lower average Ranked Probability Score ($p = 0.005$, Wilcoxon), with the advantage concentrated in low-dispersion quarters; a replication on the ECB Survey of Professional Forecasters confirms the dispersion-conditional pattern across the Euro-area panel. A dispersion-adaptive convex blend of the two methods achieves

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the lowest out-of-sample forecast score in the historical evaluation. Simulation evidence characterizes the robustness–efficiency tradeoff and identifies conditions under which each method is preferred.

Keywords: forecast combination; opinion pooling; Wasserstein distance; optimal transport; Survey of Professional Forecasters; robust aggregation

JEL codes: C53, C44, D83

1 Introduction

Combining probabilistic forecasts from multiple experts is a central problem in macroeconomics, finance, and public policy. The Survey of Professional Forecasters (SPF) elicits probability distributions over future GDP growth, inflation, and unemployment from panels of professional economists every quarter; these individual histogram forecasts must be summarized into a single aggregate distribution before informing decisions. Expert elicitation in public health, climate science, and risk analysis faces the same aggregation problem. In all of these settings the data consist of probability vectors over a finite outcome space (histogram bins, policy categories, or categorical scenarios), and the goal is to produce a single representative probability distribution.

The dominant method is the *linear opinion pool*, the arithmetic mean of the individual probability vectors [12, 14, 18]. Its appeal is axiomatic: it is the only combination rule consistent with marginalisation across subsets of outcomes [12]. In well-behaved panels with exchangeable, well-calibrated forecasters it is also asymptotically efficient. Its practical weakness is sensitivity to heterogeneous or extreme opinions. A single forecaster who places substantial mass on an extreme outcome shifts the aggregate by $1/N$ of that distance, regardless of how far outside the consensus view that forecaster sits. In the SPF, some forecasters routinely assign 30–50% probability to tail scenarios that the median forecaster assigns near zero; the arithmetic mean faithfully imports these extreme probabilities into the aggregate.

The forecast combination literature has proposed a range of alternatives to the arithmetic mean [4, 18]. The *logarithmic opinion pool* (geometric mean, renormalized) shrinks extreme probabilities because the geometric mean of a near-zero probability is near zero [12]; [17] show that *extremizing* (multiplying log-odds by a factor greater than one) sharpens aggregate predictions when experts are individually underconfident. [5] establish that nontrivially combined calibrated forecasts are in general no longer calibrated under the linear pool, and [6] study conditions under which combining predictive distributions can recover calibration. A different robustness strategy is trimming: removing the most extreme forecasters before

averaging, analogous to a trimmed mean in location estimation. Proper scoring rules provide a principled basis for evaluating probabilistic forecasts [13], and combination methods that minimize expected scoring-rule loss have been studied in the continuous-distribution setting. Other nonlinear pooling methods, such as power means, modify the influence of extreme probabilities through functional transformation; our approach instead introduces robustness through the geometry of the outcome space via optimal transport. None of these approaches offers a geometry-based robustness characterization that is directly interpretable for finite-category histogram forecasts.

Relation to existing work

The forecast-combination literature offers three main strategies for reducing sensitivity to extreme opinions in probability forecasts. The first is recalibration: [5] show that linearly combined calibrated forecasts are generally no longer calibrated, and various nonlinear pools have been proposed to restore calibration [6]. The second is trimming: removing the most extreme forecasters before averaging, studied in the SPF context by [7] and others. The third is reweighting by past accuracy, the subject of extensive theoretical and empirical work [4, 18, 8]. All three strategies are most natural when historical forecaster-level data are available for calibration or track-record estimation; in settings where such data are limited (a new panel, a rare event, a one-shot expert elicitation), a distribution-free approach is preferable.

Barycenter-based combination has recently been studied in the continuous-distribution setting. For distributions on $\mathbb{R}$, the W_2 barycenter equals the distribution whose quantile function is the component-wise mean of individual quantile functions [19]; [10] apply this to multi-model ensemble forecasting in climatology. [11] proposes depth-based trimmed means and median-like aggregates for interval and CDF forecasts, sharing our goal of robust aggregation but using a different geometric construction.

This paper makes three contributions.

Theory. We propose the Wasserstein barycenter as a robust alternative to the arithmetic mean for settings where historical reweighting by forecast accuracy is unavailable or undesirable: one-shot expert elicitations, new panels, or rare events where track records are absent. Our main theoretical result (Theorem 2.2) shows that over finite unordered outcome spaces, the Wasserstein objective simplifies to a constrained ℓ^1 problem in the probability masses—a simplification without analogue in the continuous setting that makes computation and robustness analysis tractable. For ordered outcome spaces, the W_1 barycenter has a fully closed form: the distribution whose CDF is the component-wise median of the individual CDFs (Proposition 2.3), computable without a solver. We formalize robustness via the empirical influence function and show that the barycenter's influence is bounded even as the arithmetic mean's is not.

Empirical evaluation. We validate the framework across three independent datasets spanning binary and histogram forecasts. (i) On 94 binary questions from the Good Judgment Project ($\sim$900 forecasters each), the ℓ^1 barycenter—which reduces to the component-wise median for binary outcomes—reduces mean Brier score by 12.5% relative to the arithmetic mean ($p < 0.001$). (ii) On 76 quarters of U.S. Survey of Professional Forecasters core CPI histograms (2007–2025), the W_1 barycenter achieves a lower mean Ranked Probability Score ($p = 0.005$, Wilcoxon) and places more probability mass on the realized outcome in a majority of quarters; the advantage is concentrated in low-dispersion quarters where the panel is broadly in agreement. (iii) A replication on the ECB Survey of Professional Forecasters Euro-area HICP histograms confirms the dispersion-conditional pattern on an independent international panel. A secondary opioid-survey application illustrates the unordered case with $K = 20$ outcomes [9].

Adaptive blending. A convex blend of the barycenter and the arithmetic mean—with weight estimated on an expanding historical window—achieves the lowest out-of-sample Ranked Probability Score, improving over the pure barycenter by 1.7% and over the arith-

metic mean by 4.5%. The optimal mixture weight ($\lambda^* \approx 0.70$) is stable across time, indicating that the gain is not sensitive to the choice of training window.

The remainder of the paper is organized as follows. Section 2 states the two main theoretical results: the ℓ^1 simplification theorem for unordered spaces and the CDF-median characterization for ordered spaces. Section 3 formalizes robustness via the empirical influence function. Section 4 reports simulation evidence. Section 5 presents the opioid expert survey (unordered outcomes, $K = 20$) and a binary scored evaluation using the Good Judgment Project. Section 6 presents the U.S. SPF application: single-quarter illustration, 76-quarter historical backtest, dispersion-adaptive blend, and ECB SPF replication. Section 7 concludes.

2 Wasserstein Barycenters as Aggregate Measures

Let $\Omega = \{1, \ldots, K\}$ be a finite set of outcomes. Write $P \in \mathbb{R}^{N \times K}$ for the matrix of expert probability vectors, with row $P_{j\cdot} \in \Delta_{K-1}$ representing expert j's distribution, and let D be the $K \times K$ matrix of ground distances between outcomes. Because Δ_{K-1} is compact and the Wasserstein objective is continuous, a barycenter exists for any ground cost [19]. The main result of this section is that when D is the indicator distance, the barycenter objective simplifies to an ℓ^1 problem in the probability masses (Theorem 2.2), giving a computationally tractable and geometrically transparent characterization. The ordered case ($d(x, y) = |x - y|$) yields a further closed-form result (Proposition 2.3).

2.1 Definition of the Wasserstein Barycenter

The following definition can be found in [21]. Let (Ω, d) be a metric space for which every measure is a Radon measure and let $\mathcal{P}(\omega)$ be the set of measures on Ω with finite p^{th} moment for some element in Ω.

Definition 1. *A Wasserstein barycenter of N measures $\{P_1, \ldots, P_N\}$ is a minimizer of*

$$f(\Psi) = \frac{1}{N} \sum_{i=1}^{N} W_p^p(\Psi, P_i).$$

over the set of all measures over the state space Ω where W_p is the Wasserstein distance given by

$$W_p = \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int d(x,y)^p d\pi(x,y) \right)^{1/p}.$$

The authors of [20] present results for the support of discrete Wasserstein barycenter for finitely supported measures. That is, each measure places mass on a finite number of points in Euclidean space. The authors show the set of support points is not contained in the union of the support of the individual measures. Hence discrete Wasserstein barycenters in Euclidean space are not bounded by the component probability measures. But, we are concerned with the case such that $\Omega = \{1, \ldots, K\}$. The definition becomes any Ψ that minimizes

$$\sum_{j=1}^{N} \left(\min_{\pi \in \Gamma(p_j, \Psi)} \sum_{x=1}^{K} \sum_{y=1}^{K} d(x,y)^p \pi(x,y) \right).$$

We consider two ground costs: the indicator distance for unordered outcome spaces (Section 2.2) and the absolute-value distance $d(x,y) = |x - y|$ for ordered spaces.

2.2 Unordered Finite Outcome Space

When the distance function is given by

$$d(x,y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

choose matrices $\Pi^1, \ldots, \Pi^N$ to minimize the objective function

$$
\begin{aligned}
f(\mathbf{\Pi}) &= \sum_{l=1}^{N} \left(\sum_{k=1}^{K} \sum_{k'=1}^{K} d(k,k')^2 \Pi^l_{k,k'} \right) \\
&= \sum_{l=1}^{N} \sum_{k=1}^{K} \sum_{k' \neq k} \Pi^l_{k,k'}
\end{aligned}
$$

subject to the feasibility conditions for $j = 1, \ldots, N$ and $l, m = 1, \ldots, N$

$$
\begin{aligned}
\sum_j \Pi^l_{k,j} &= p_l(k) \\
\sum_j \Pi^l_{j,k} &= \sum_j \Pi^m_{j,k}.
\end{aligned}
$$

Note that the objective function is the sum of the off-diagonal probabilities,

$$
\sum_k \sum_{k' \neq k} \Pi^l(k,k') = 1 - \sum_k \Pi^l(k,k).
$$

Hence minimizing the objective function is equivalent to maximizing the probability mass placed on the diagonal entries of the joint probability density matrices $\{\Pi^l\}$. The following lemma shows that this maximum is achieved by every optimal transport plan, and identifies the value of the diagonal entries.

Lemma 2.1 (Indicator-cost OT)**.** *Let $p, \Psi \in \Delta_{K-1}$. Under the indicator cost, every optimal transport plan $\Pi^* \in \Gamma(p, \Psi)$ satisfies*

$$
\Pi^*_{kk} = \min\{p(k),\ \Psi(k)\} \quad \textit{for all } k.
$$

Proof. Since $d(k,k') = 0$ when $k = k'$ and $d(k,k') = 1$ otherwise, the transport cost equals $1 - \sum_k \Pi_{kk}$, so any optimal plan maximises $\sum_k \Pi_{kk}$. For any feasible $\Pi \in \Gamma(p, \Psi)$ the marginal constraints give $\sum_{k'} \Pi_{kk'} = p(k)$ and $\sum_{k'} \Pi_{k'k} = \Psi(k)$, so $\Pi_{kk} \leq \min\{p(k), \Psi(k)\}$. The plan $\Pi^*_{kk} = \min\{p(k), \Psi(k)\}$ attains the upper bound. Feasibility requires the residual row mass

$p(k) - \Pi^*_{kk} = \max\{p(k) - \Psi(k), 0\}$ and residual column mass $\Psi(k) - \Pi^*_{kk} = \max\{\Psi(k) - p(k), 0\}$ to be assigned to off-diagonal entries. Both residuals are nonnegative and their totals across k are equal (both equal $\sum_k \max\{p(k) - \Psi(k), 0\}$, since p and Ψ are probability vectors), so a feasible assignment of the residuals to off-diagonal entries exists by a standard flow argument. □

The analogous property for barycenter solutions is established in Appendix B: every optimal plan Π^l simultaneously satisfies $\Pi^l_{kk} = \min\{p_l(k), \Psi(k)\}$.

Theorem 2.2 (ℓ^1 Simplification)**.** *Let $\{p_1, \ldots, p_N\}$ be probability mass functions on $\{1, \ldots, K\}$ and let the ground cost be the indicator distance. Then the Wasserstein barycenter Ψ^* solves*

$$\min_{\Psi \in \Delta_{K-1}} \sum_{l=1}^{N} \sum_{k=1}^{K} |p_l(k) - \Psi(k)| \,, \tag{2.1}$$

the ℓ^1 sum of componentwise deviations from the experts. In particular, a Wasserstein barycenter exists and the Wasserstein objective is equivalent to an ℓ^1 regression of Ψ onto the expert opinions.

Proof. Fix any candidate $\Psi \in \Delta_{K-1}$. By Lemma 2.1, the optimal transport cost $\mathrm{OT}(\Psi, p_l)$ between Ψ and each p_l under the indicator distance equals

$$\mathrm{OT}(\Psi, p_l) = 1 - \sum_k \min\{p_l(k), \Psi(k)\} = \tfrac{1}{2} \sum_k |p_l(k) - \Psi(k)|,$$

using the identity $a - \min(a, b) = \max(a - b, 0)$ and the fact that $\sum_k (p_l(k) - \Psi(k)) = 0$. Summing over l gives the objective (2.1) up to a positive constant. The function $\Psi \mapsto \sum_{l,k} |p_l(k) - \Psi(k)|$ is continuous on the compact simplex, so the minimum is attained. □

Remark 1. *The identity $\mathrm{OT}(p, \Psi) = \frac{1}{2}\|p - \Psi\|_1$ is the standard equivalence between the Wasserstein-1 distance under the indicator cost and the total variation distance; see [16] for a survey. The theorem reduces the barycenter problem to ℓ^1 median estimation on the*

simplex: the unconstrained minimizer assigns each state the coordinatewise median of the expert opinions, and the simplex constraint shifts this toward a normalized version of the median. The arithmetic mean generally does not satisfy this condition and therefore does not generally coincide with the Wasserstein barycenter.

2.3 Discrete Barycenters for Ordered Spaces

When the outcome space $\Omega = \{1, \ldots, K\}$ carries a natural ordering, the indicator distance used in the preceding section discards structural information. Setting $d(x, y) = |x - y|$ makes (Ω, d) a metric space that respects the ordering and gives rise to a distinct class of Wasserstein barycenters. Let $F_j(k) = \sum_{x \leq k} P_j(x)$ denote the cumulative distribution function (CDF) of P_j and let $F_j^{-1}(t) = \inf\{k : F_j(k) \geq t\}$ denote its quantile function.

Wasserstein-1 Barycenters

For $p = 1$, the Wasserstein distance between P and Q on Ω reduces to the L^1 distance between their CDFs [15, 16]:

$$W_1(P, Q) = \sum_{k=1}^{K-1} |F_P(k) - F_Q(k)| \,.$$

The barycenter minimizes $\sum_{j=1}^{N} W_1(\Psi, P_j)$ over probability mass functions Ψ on Ω. Substituting the CDF representation and exchanging the order of summation gives

$$\sum_{j=1}^{N} W_1(\Psi, P_j) = \sum_{k=1}^{K-1} \sum_{j=1}^{N} |F_\Psi(k) - F_j(k)| \,,$$

which separates across k. Each term is independently minimized by the sample median of $\{F_j(k)\}_{j=1}^{N}$.

Proposition 2.3. *Let $\{P_1, \ldots, P_N\}$ be probability mass functions on $\{1, \ldots, K\}$ and let W_1*

be defined with $d(x, y) = |x - y|$. A Wasserstein-1 barycenter has CDF

$$F_\Psi(k) = \operatorname{med}\{F_1(k), \ldots, F_N(k)\}, \quad k = 1, \ldots, K-1.$$

Proof. Since the objective separates across k, the unconstrained minimizer at each k is the sample median of $\{F_j(k)\}_{j=1}^N$. It remains to verify that the vector of component-wise medians constitutes a valid CDF, i.e., it is nondecreasing in k. Since each F_j is nondecreasing, $F_j(k) \leq F_j(k+1)$ for every j. If $x_j \leq y_j$ for all j, then $\operatorname{med}\{x_j\} \leq \operatorname{med}\{y_j\}$; hence $F_\Psi(k) \leq F_\Psi(k+1)$ for all k. The boundary conditions $F_\Psi(0) = 0$ and $F_\Psi(K) = 1$ hold since $F_j(0) = 0$ and $F_j(K) = 1$ for every j. Thus F_Ψ is a valid CDF and $\Psi(k) = F_\Psi(k) - F_\Psi(k-1) \geq 0$ defines a probability mass function on Ω. □

The W-1 barycenter has a natural interpretation: it is the distribution whose CDF lies at the "center" of the collection in the L^1 sense. Boundedness of Ψ in the CDF sense is immediate since the median is bounded by the minimum and maximum: $\min_j F_j(k) \leq F_\Psi(k) \leq \max_j F_j(k)$ for all k.

3 Robustness of Aggregating Measures

Both the arithmetic mean and the Wasserstein barycenter are bounded aggregating measures, as established in the preceding sections. They differ, however, in their sensitivity to extreme or outlying expert opinions. This section formalizes that difference via the empirical influence function and illustrates it through simulation.

3.1 Empirical Influence Function

The empirical influence function (EIF) quantifies the effect of introducing a single additional expert P_0 into an existing collection $\mathbf{P}_N = \{P_1, \ldots, P_N\}$. For an aggregating measure T, the

EIF is

$$\mathrm{EIF}(P_0;\, T,\, \mathbf{P}_N) = (N+1)\left[T(\mathbf{P}_N \cup \{P_0\}) - T(\mathbf{P}_N)\right],$$

rescaled so that it approximates the derivative of T with respect to contamination fraction $\varepsilon = 1/(N+1)$. The gross error sensitivity $\mathrm{GES}(T) = \sup_{P_0} \|\mathrm{EIF}(P_0;\, T,\, \mathbf{P}_N)\|_\infty$ summarizes the worst-case influence of a single outlier.

Proposition 3.1 (**EIF of the Arithmetic Mean**). *Let $\bar{P}_N = \frac{1}{N}\sum_{j=1}^{N} P_j$ be the arithmetic mean of $\mathbf{P}_N$. Then*

$$\mathrm{EIF}_k(P_0;\, \bar{P},\, \mathbf{P}_N) = P_0(k) - \bar{P}_N(k)$$

for each state $k \in \Omega$. The gross error sensitivity is $\mathrm{GES}(\bar{P}) = \sup_{P_0} \left\|P_0 - \bar{P}_N\right\|_\infty$, which is maximized as $P_0 \to \delta_k$ for any k, giving $\mathrm{GES}(\bar{P}) \leq 1 - \min_k \bar{P}_N(k)$.

Proof. Adding P_0 to $\mathbf{P}_N$ gives $\bar{P}_{N+1}(k) = \frac{N\bar{P}_N(k)+P_0(k)}{N+1}$. Then

$$(N+1)\left[\bar{P}_{N+1}(k) - \bar{P}_N(k)\right] = (N+1)\cdot\frac{P_0(k) - \bar{P}_N(k)}{N+1} = P_0(k) - \bar{P}_N(k).$$

The EIF is linear in P_0 and equals zero only when $P_0 = \bar{P}_N$. As $P_0(k) \to 1$ for any k, the influence approaches $1 - \bar{P}_N(k)$, which is bounded away from zero whenever $K > 1$. □

For the Wasserstein barycenter, the EIF depends on the geometry of the Wasserstein space. Adding P_0 introduces a new term $\frac{1}{N+1}W_p^p(\Psi, P_0)$ into the barycenter objective, which must be balanced against the $\frac{1}{N+1}$ reduction in weight on each of the N existing terms. The new minimizer $\hat{\Psi}_{N+1}$ shifts from $\hat{\Psi}_N$ only to the extent that moving toward P_0 does not excessively increase the total Wasserstein cost relative to the existing N experts.

Proposition 3.2 (**EIF of the Wasserstein Barycenter**). *Let $\hat{\Psi}_N$ and $\hat{\Psi}_{N+1}$ denote the Wasserstein barycenters of $\mathbf{P}_N$ and $\mathbf{P}_N \cup \{P_0\}$ respectively. The shift $\hat{\Psi}_{N+1}(k) - \hat{\Psi}_N(k)$ is*

bounded by the range of expert opinions on state k:

$$\left|\hat{\Psi}_{N+1}(k) - \hat{\Psi}_N(k)\right| \leq \max\left(\max_{j\leq N} P_j(k),\ P_0(k)\right) - \min\left(\min_{j\leq N} P_j(k),\ P_0(k)\right).$$

Proof. Both $\hat{\Psi}_N(k)$ and $\hat{\Psi}_{N+1}(k)$ lie within the range of their respective expert collections (Theorem 2.2, boundedness), so their difference cannot exceed the range of the combined collection. □

The two propositions reveal a structural asymmetry. The EIF of the arithmetic mean equals $P_0 - \bar{P}_N$ exactly, independent of N: every expert, regardless of how extreme, exerts unit influence on the aggregate. The barycenter's shift is bounded by the range of expert opinions; an outlying opinion far from the existing consensus is costly in the Wasserstein sense and therefore exerts reduced influence. This is median-like in spirit: the barycenter's response is constrained by the geometry of the existing expert panel rather than being a linear function of all views. We do not claim a full robust-statistics treatment with sharp breakdown-point results; rather, the bound provides an interpretable upper limit on how much any single opinion can shift the aggregate.

3.2 Simulation Evidence

To illustrate the practical difference, we simulate $R = 200$ collections of $N = 11$ experts. Ten experts are drawn from a uniform Dirichlet distribution over $K = 6$ states. One expert is drawn from a concentrated Dirichlet placing substantial mass on a single state (the "outlier"). For each collection, both the arithmetic mean and the Wasserstein barycenter are computed.

Figure 1 displays the distribution of the probability assigned to the outlier's preferred state (S_6) across the R replicates. The arithmetic mean is substantially more inflated: in 90% of replicates the AM assigns more probability to S_6 than the barycenter does, consistent with its equal-weight treatment of the outlier. Figure 2 shows the variance of each aggregate across replicates by state. The barycenter has lower variance on the outlier state (0.0039 vs.

0.0016 for the AM on S_6) but higher overall variance, confirming the classical robustness–efficiency tradeoff.

Figure 3 displays a single representative draw: five regular experts, one outlier, and the two resulting aggregates. The arithmetic mean is visibly pulled toward the outlier's distribution; the barycenter remains closer to the consensus of the regular experts.

Figure 4 varies the number of outlying experts from zero to three. The arithmetic mean's assignment to S_6 grows approximately linearly with outlier count. The barycenter grows more slowly, reflecting its distributional resistance.

Remark 2. *The higher variance of the barycenter across replicates is expected. When expert opinions are drawn from a well-specified population, the arithmetic mean is the minimum-variance unbiased estimator of the population mean. The barycenter trades efficiency for robustness, analogous to the median relative to the mean in location estimation. The practical choice between the two depends on the degree of belief in the homogeneity of the expert pool.*

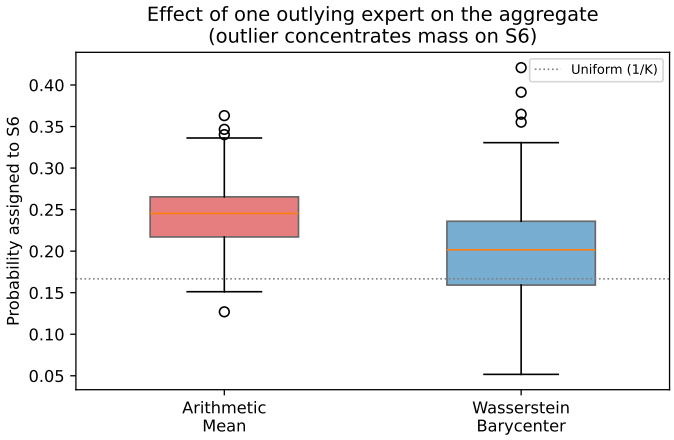

Figure 1: Distribution of the probability assigned to the outlier's preferred state (S_6) across $R = 200$ simulation replicates. The arithmetic mean (red) is systematically more inflated than the Wasserstein barycenter (blue).

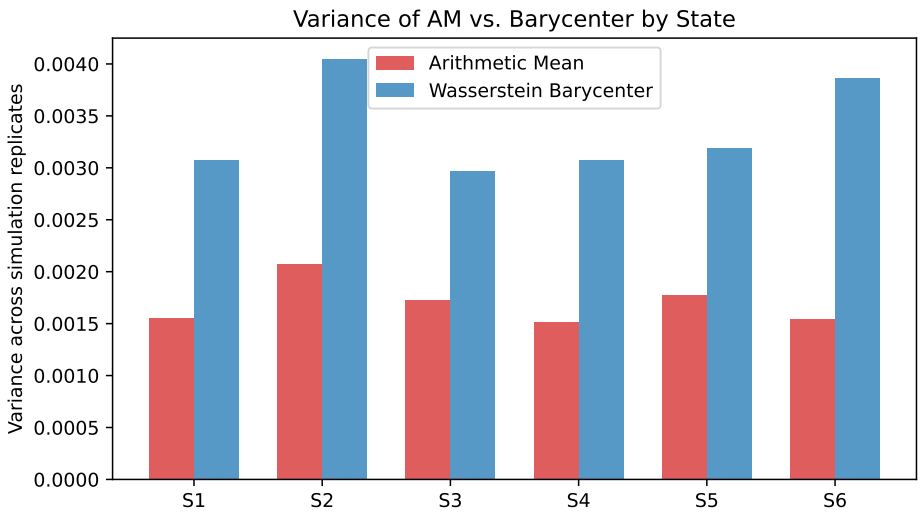

Figure 2: Variance of the arithmetic mean and Wasserstein barycenter by state across $R = 200$ replicates. The barycenter has lower variance on the outlier state (S_6) but higher variance on regular states, illustrating the robustness–efficiency tradeoff.

4 Simulations

The robustness analysis of the preceding section held the structure of contamination fixed: one outlying expert drawn from a concentrated Dirichlet, $N-1$ regular experts drawn from a uniform Dirichlet. This section broadens the comparison through a systematic simulation study examining (1) efficiency under a correctly specified population model and (2) robustness under varying degrees of misspecification.

4.1 Setup

All simulations use $K = 6$ states. Expert opinions are drawn from one of two Dirichlet populations:

- **Regular experts**: $P_j \sim \mathrm{Dir}(\mathbf{1}_K)$, the uniform Dirichlet placing equal prior weight on all states.
- **Outlying experts**: $P_j \sim \mathrm{Dir}(\alpha_{\text{out}})$ with $\alpha_{\text{out}} = (1, 1, 1, 1, 1.5, 10)^{\odot 4}$, a highly concentrated distribution placing substantial mass on state S_6.

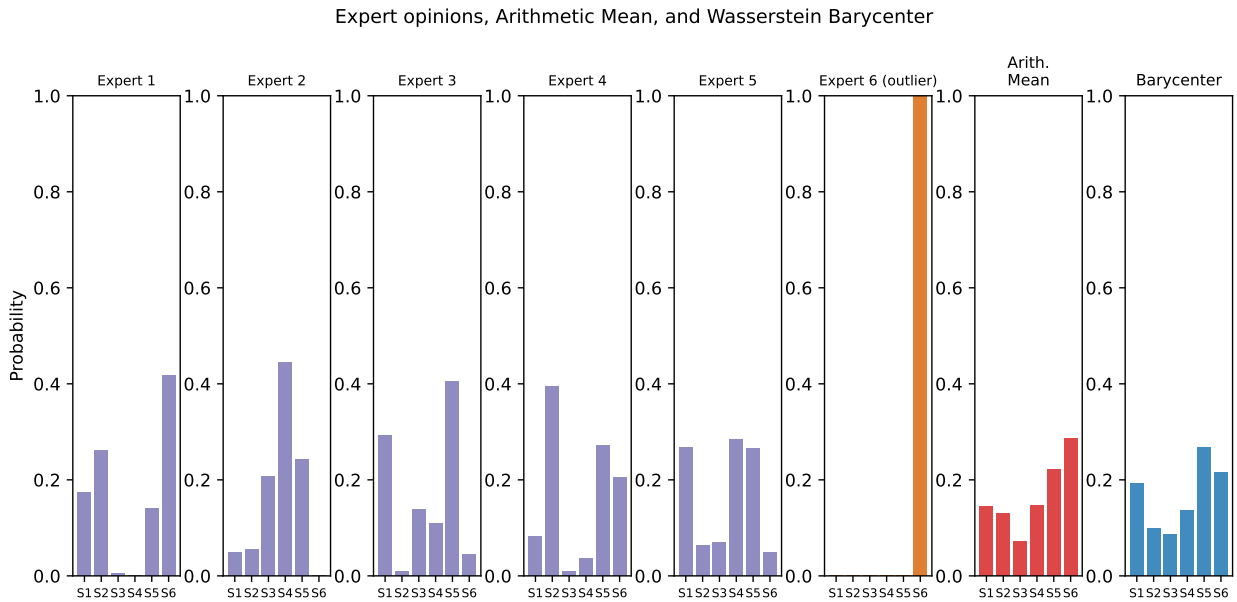

Figure 3: A single simulation draw: five regular experts (purple), one outlier (orange), arithmetic mean (red), and Wasserstein barycenter (blue). The arithmetic mean is visibly pulled toward the outlier's distribution.

For each simulation, N_{reg} regular and N_{out} outlying opinions are drawn; the arithmetic mean and Wasserstein barycenter (with uniform weights and indicator distance) are computed; and the results are summarized across $R = 200$ replicates. Performance is measured by (i) mean squared error relative to the population mean of the regular experts, (ii) variance across replicates, and (iii) probability assigned to the outlier's preferred state S_6.

4.2 Efficiency Under the Null

When all experts are drawn from the same regular population ($N_{\text{out}} = 0$), the arithmetic mean is the minimum-variance unbiased estimator of the population probability vector. Table 1 compares the mean squared error of the two aggregates relative to the true population mean $\mu = \mathbf{1}_K/K$ for varying N.

Both estimators are consistent: MSE decreases at rate $O(1/N)$. The barycenter's efficiency loss relative to the AM grows with N, reaching a ratio of 1.86 at $N = 20$. This reflects the L^1 vs. L^2 character of the two estimators: as more data arrives, the AM concen-

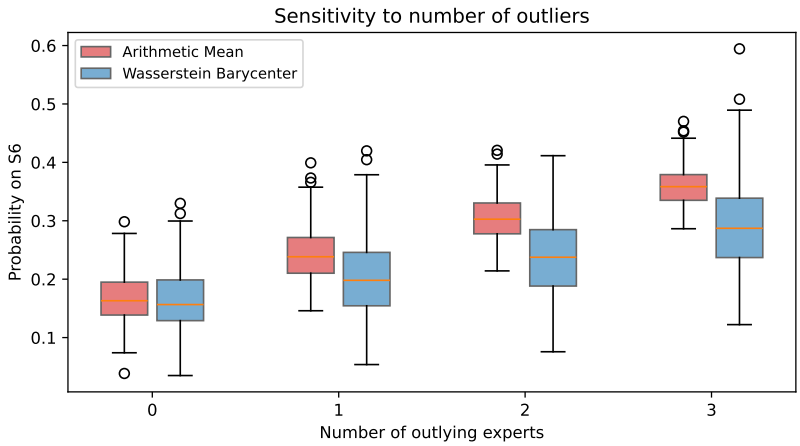

Figure 4: Sensitivity to outlier count. As the number of outlying experts increases from zero to three, the arithmetic mean's assignment to S_6 grows more rapidly than the barycenter's.

N	AM MSE	BC MSE	BC/AM ratio
5	0.0225	0.0317	1.41
10	0.0124	0.0198	1.59
20	0.0063	0.0118	1.86

Table 1: Mean squared error (summed over states) of the arithmetic mean and Wasserstein barycenter relative to the true population mean $\mu = \mathbf{1}_K/K$, averaged over $R = 200$ replicates with no outlying experts. The AM is more efficient in all cases; the ratio grows with N.

trates its distribution more tightly, while the barycenter's Wasserstein objective introduces a systematic dispersion penalty.

4.3 Robustness Under Contamination

Figure 4 (Section 3) already showed that the barycenter resists outliers more effectively as their count increases from zero to three. Here we quantify the tradeoff more precisely. For $N_{\text{reg}} = 10$ and $N_{\text{out}} \in \{0, 1, 2, 3\}$, Table 2 reports the mean and standard deviation of the probability assigned to S_6 under each aggregate.

The arithmetic mean's assignment to S_6 increases by $\approx 1/N_{\text{tot}}$ per additional outlier, consistent with the EIF derived in Section 3. The barycenter's increase is approximately $1.7\times$ smaller, and its standard deviation rises modestly as expert opinions become more heterogeneous, while the AM standard deviation falls (the outlier deterministically anchors

N_{out}	AM mean (S_6)	AM sd	BC mean (S_6)	BC sd
0	0.165	0.044	0.166	0.057
1	0.243	0.041	0.202	0.061
2	0.309	0.040	0.245	0.075
3	0.357	0.033	0.281	0.074

Table 2: Mean and standard deviation of the probability assigned to state S_6 as the number of outlying experts increases. The AM mean grows by approximately 0.064 per additional outlier; the BC mean grows by approximately 0.038.

the mean toward S_6).

4.4 Sensitivity to Population Model

A natural question is how performance depends on the degree of heterogeneity in the regular expert population. We vary the Dirichlet concentration parameter α_0 over a log-spaced grid from 0.25 to 20 (with $\alpha = \alpha_0 \mathbf{1}_K$), where smaller values produce more dispersed expert opinions. Figure 5 shows that the MSE ratio decreases monotonically as α_0 increases, falling from 2.45 at $\alpha_0 = 0.25$ to 1.35 at $\alpha_0 = 20$. When experts are highly dispersed (small α_0), the barycenter's ℓ^1 objective assigns heavier weight to large deviations, amplifying its efficiency loss relative to the ℓ^2-optimal AM. As the population concentrates, expert opinions converge and the two aggregates become more similar.

5 Application to a Survey of Public Health Experts

We apply the methods developed above to a survey conducted by the *New York Times* in which 30 public health experts were each asked to allocate a hypothetical budget across 20 policy interventions aimed at addressing the opioid crisis. Each expert's allocation is a probability vector over the 20 interventions and the collection of 30 vectors constitutes a natural instance of the expert aggregation problem.

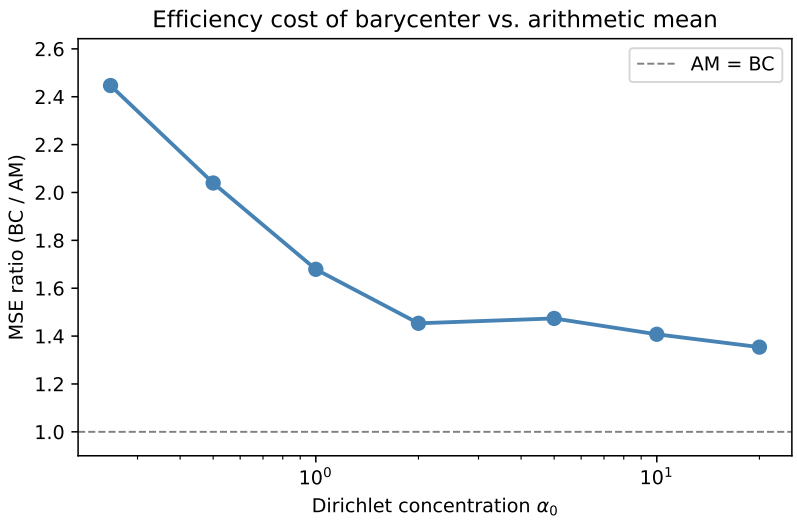

Figure 5: MSE ratio (BC/AM) as a function of the Dirichlet concentration α_0, with $N = 10$ regular experts and no outliers. The ratio is stable across population models.

5.1 Data

The 20 interventions span four policy categories: *treatment* (medication-assisted treatment, medicaid expansion, addiction treatment in prisons, drug courts, research); *harm reduction* (naloxone, needle exchanges, supervised consumption spaces, HIV/HCV treatment, fentanyl testing strips, medical examiners); *prevention* (community development, education, pain research, post-incarceration social programs); and *supply* (interdiction, prescription drug monitoring, local police forces, border wall, reducing diversion). Expert allocations range from near-zero (many experts allocate nothing to border-wall or supervised consumption spaces) to over 50% (one expert allocates half their budget to medication-assisted treatment alone). Figure 6 displays the full allocation matrix.

5.2 Aggregate Distributions

Figure 7 shows the arithmetic mean and Wasserstein barycenter side by side. Both aggregates identify medication-assisted treatment as the top priority ($\bar{P} = 0.183$, $\hat{\Psi} = 0.195$), followed by community development, medicaid expansion, and addiction treatment in prisons. The two aggregates agree on the ranking of the top interventions but differ systematically in their

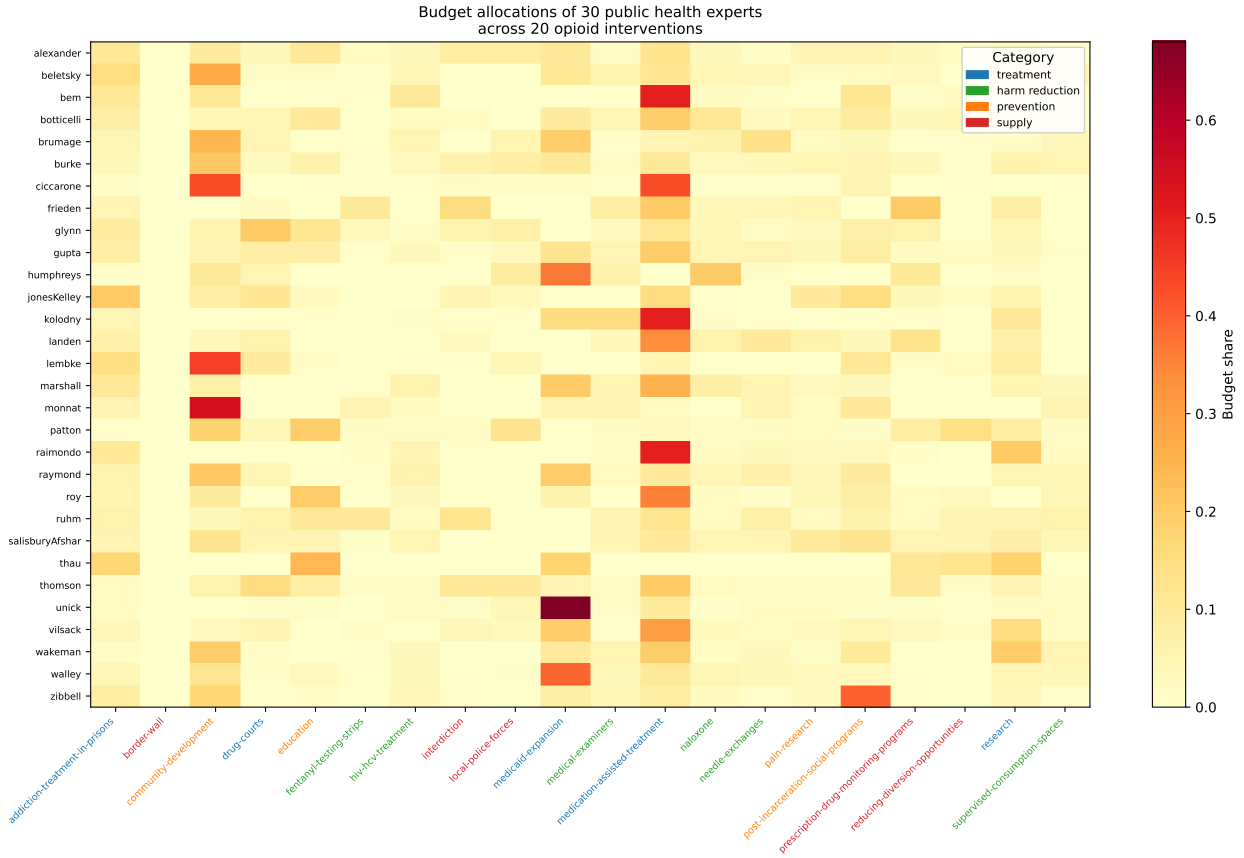

Figure 6: Budget allocations of 30 public health experts across 20 opioid interventions. Cell color encodes the budget share; intervention labels are colored by policy category (blue = treatment, green = harm reduction, orange = prevention, red = supply).

category-level allocation.

Table 3 reports the total budget share assigned to each category. The barycenter allocates more to treatment (0.481 vs. 0.468) and harm reduction (0.173 vs. 0.155) and less to prevention (0.262 vs. 0.270) and supply (0.084 vs. 0.107) than the arithmetic mean. The largest individual-intervention differences are in community development ($\bar{P} - \hat{\Psi} = +0.013$), medicaid expansion ($+0.012$), and medication-assisted treatment (-0.012).

5.3 Leave-One-Out Influence

To assess the influence of individual experts, we compute the ℓ^1 shift in each aggregate when each expert is removed in turn. This is the empirical counterpart of the EIF studied in

Category	Arithmetic Mean	Barycenter
Treatment	0.468	0.481
Harm reduction	0.155	0.173
Prevention	0.270	0.262
Supply	0.107	0.084

Table 3: Total budget share allocated to each policy category by the arithmetic mean and Wasserstein barycenter.

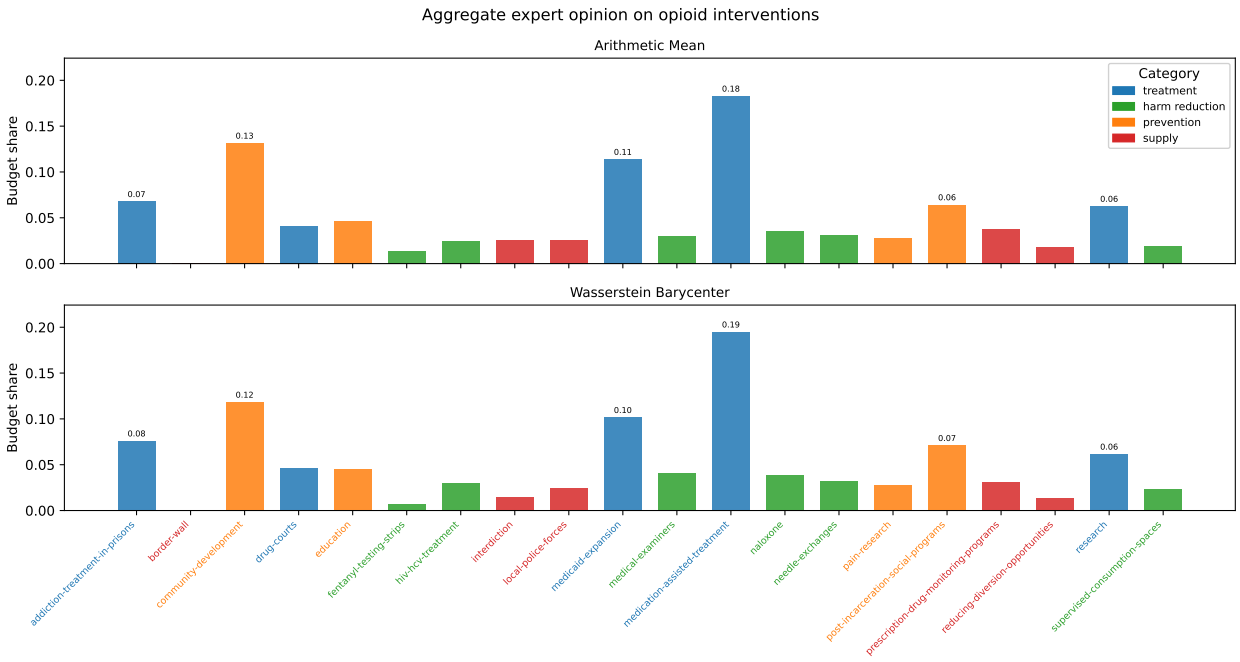

Figure 7: Arithmetic mean (top) and Wasserstein barycenter (bottom) of the 30 expert allocations. Bars are colored by policy category. Annotations show allocations exceeding 5%.

Section 3. Figure 8 displays the leave-one-out influence of each expert, ordered by their influence on the arithmetic mean.

The most influential experts on the arithmetic mean are those whose allocations deviate most from the consensus, concentrating nearly all of their budget on a single intervention. For this dataset, the leave-one-out shifts are comparable in magnitude for the two methods; mean shifts are 2.7 percentage points for the AM and 3.6 for the barycenter. This is consistent with the data structure: without a clearly outlying expert, both aggregators respond similarly to removals. The robustness advantage of the barycenter is most pronounced when a minority

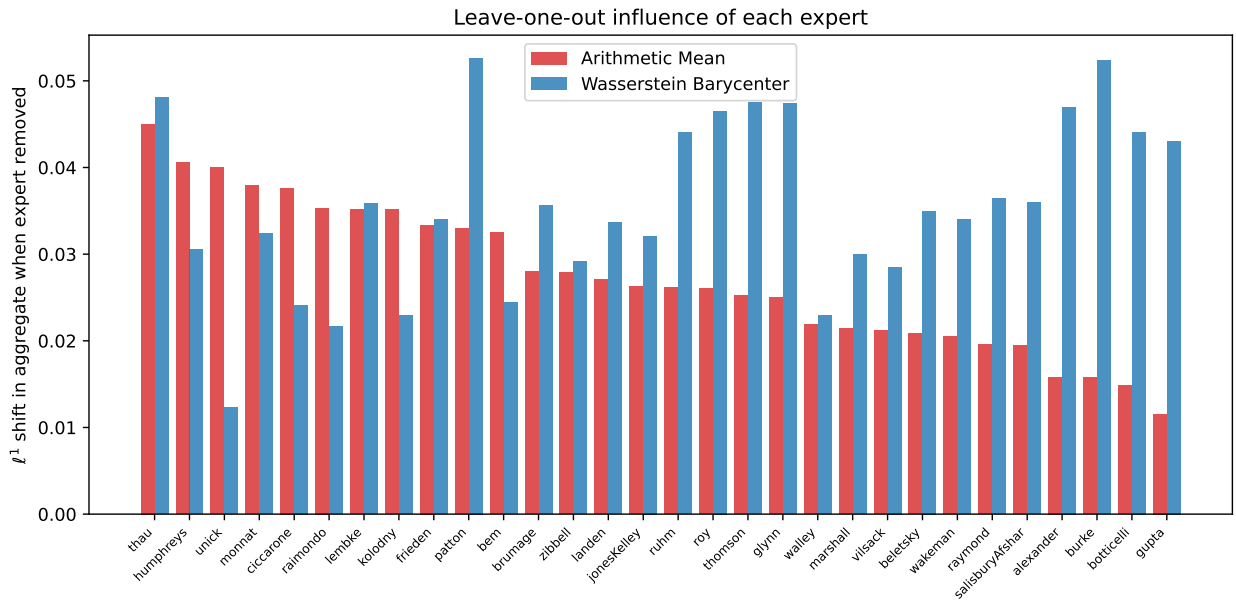

Figure 8: Leave-one-out ℓ^1 shift in the arithmetic mean (red) and Wasserstein barycenter (blue) when each expert is removed. Experts are ordered by their influence on the arithmetic mean.

of experts holds extreme views, as in the SPF application (Section 6) and the contamination simulations (Section 4). Figure 9 further illustrates where the two methods agree and disagree at the intervention level.

5.4 Historical Evaluation: Good Judgment Project

The opioid application illustrates the unordered barycenter with $K = 20$ outcomes but lacks verified realizations for scoring. We complement it with a scored evaluation on binary ($K = 2$) questions from the Good Judgment Project (GJP), a geopolitical forecasting tournament conducted under the IARPA ACE program from 2011 to 2015 [2].

For binary outcomes, the indicator-cost ℓ^1 barycenter reduces to the component-wise median of the individual probability vectors. Each forecaster's distribution over {Yes, No} is characterized by a single number $p_j \in [0, 1]$; the barycenter selects $p^* = \text{median}_j(p_j)$ while the arithmetic mean selects $\bar{p} = \frac{1}{N}\sum_j p_j$. The dispersion-robustness intuition is direct: if a minority of forecasters assigns extreme probabilities near 0 or 1, the median is unaffected

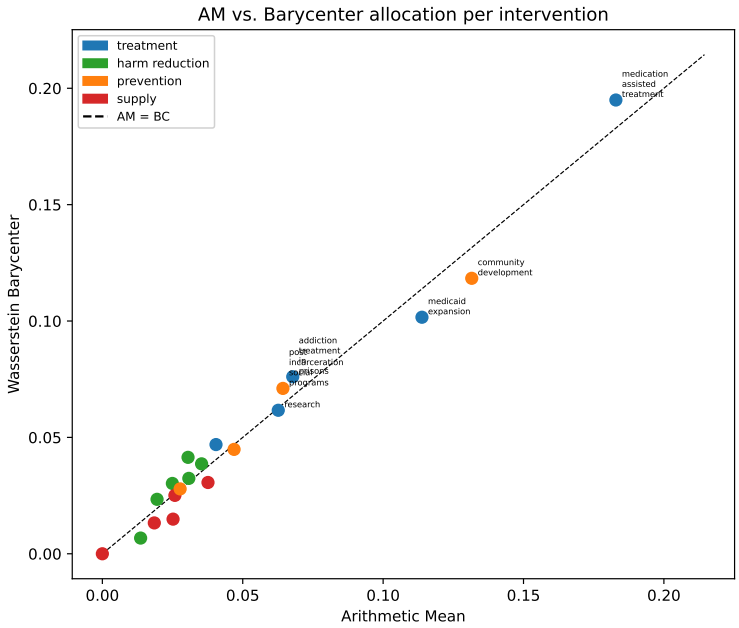

Figure 9: AM vs. barycenter allocation for each of the 20 interventions. Points above the diagonal indicate interventions where the barycenter allocates more than the AM; points below indicate the reverse. Colors denote policy category.

while the mean is pulled toward them.

Data and protocol. We use individual survey forecasts from Year 1 of the GJP (2011–2012), publicly available through Harvard Dataverse [3]. For each of the 94 binary questions with at least 5 participating forecasters (mean panel size: 895), we take each forecaster's last submitted forecast before question closure and compute the arithmetic mean and median of the individual $P(\text{Yes})$ values. We score each aggregate against the binary resolution using the Brier score, $\text{BS} = (p - \mathbf{1}_{\text{Yes}})^2$.

Results. The median (barycenter) achieves a lower Brier score than the arithmetic mean on 85.1% of questions, with mean Brier scores of 0.091 (BC) versus 0.104 (AM), a 12.5% improvement ($p < 0.001$, Wilcoxon). The advantage is present across all dispersion terciles and is largest when the panel exhibits low disagreement (Table 4 and Figure 10).

The GJP finding adds an important dimension: the barycenter's advantage holds in a setting where the number of forecasters is very large ($\sim$900 per question) and outcomes are

Dispersion group	AM Brier	BC Brier	BC improvement
Low tercile	0.034	0.018	45.1%
Mid tercile	0.123	0.107	13.0%
High tercile	0.157	0.149	5.3%
Overall	0.104	0.091	12.5%

Table 4: GJP Year 1 Brier scores by dispersion tercile ($n = 94$ binary questions, mean panel size 895). The barycenter (= component-wise median) outperforms the arithmetic mean most strongly when the panel is cohesive.

binary. Even with nearly a thousand forecasters, the median dominates the mean because it is robust to the minority of participants who assign extreme probabilities.

6 Application to the Survey of Professional Forecasters

The opioid survey treats the 20 interventions as an *unordered* outcome space, so the indicator distance is the natural choice. Many expert-aggregation problems, however, involve *ordered* outcomes: survey respondents assign probabilities to ordered bins. For such data the W_1 barycenter has the closed-form characterization derived in Proposition 2.3: it equals the component-wise median of the empirical CDFs. This section illustrates both the practical convenience of that formula and its robustness implications using data from the Survey of Professional Forecasters (SPF).

6.1 Data

The Philadelphia Federal Reserve Bank has administered the SPF every quarter since 1968. In each survey, professional forecasters submit probabilistic predictions for macroeconomic variables by allocating probability mass across a fixed set of ordered bins. We use the *core CPI* probability question (variable `PRCCPI`), which asks each forecaster to assign probabilities to 10 ordered bins for fourth-quarter-over-fourth-quarter core CPI inflation in the current and following year. The 10 bins are (in ascending order): decline, 0.0–0.4%, 0.5–0.9%, 1.0–1.4%, 1.5–1.9%, 2.0–2.4%, 2.5–2.9%, 3.0–3.4%, 3.5–3.9%, and $\geq$ 4.0%.

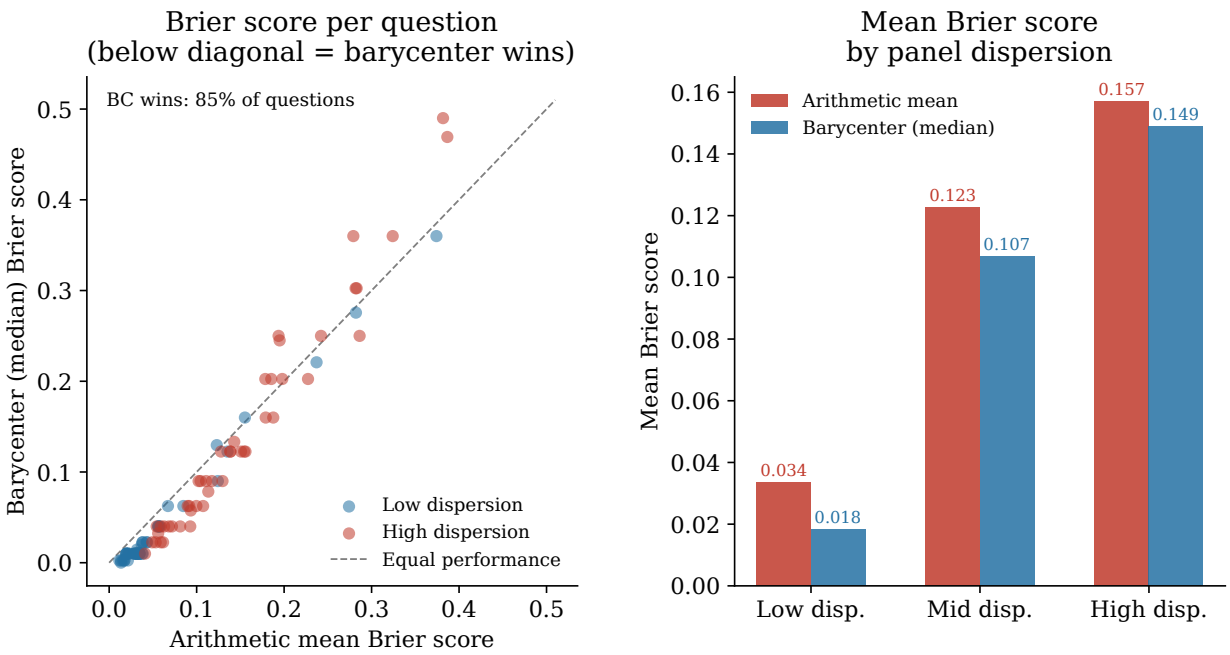

Figure 10: GJP Year 1: Brier score comparison for 94 binary questions. Left: scatter of AM vs. BC Brier score (one point per question; below the diagonal = BC wins; color indicates panel dispersion). Right: mean Brier score by dispersion tercile. The barycenter consistently outperforms the arithmetic mean.

We focus on the 2022:Q1 survey's *next-year* histograms, each forecaster's probability distribution for 2023 core CPI inflation, using the $n = 27$ respondents who provided a complete histogram summing to 100%. This survey was conducted in February 2022, when realized core CPI was running near 6% and forecasters disagreed sharply about how quickly inflation would moderate.

6.2 Aggregate Distributions

Figure 11 displays the 27 individual histograms as a heatmap. The heterogeneity is striking: some forecasters concentrated nearly all mass on 2–3% (expecting a rapid return to target), while others spread mass across higher bins. A few placed non-trivial probability on the decline and $\geq 4\%$ tail bins.

Because the outcomes are ordered, we use the W_1 distance (absolute rank difference) as the ground cost. This respects the ordinal structure of the inflation bins: shifting prob-

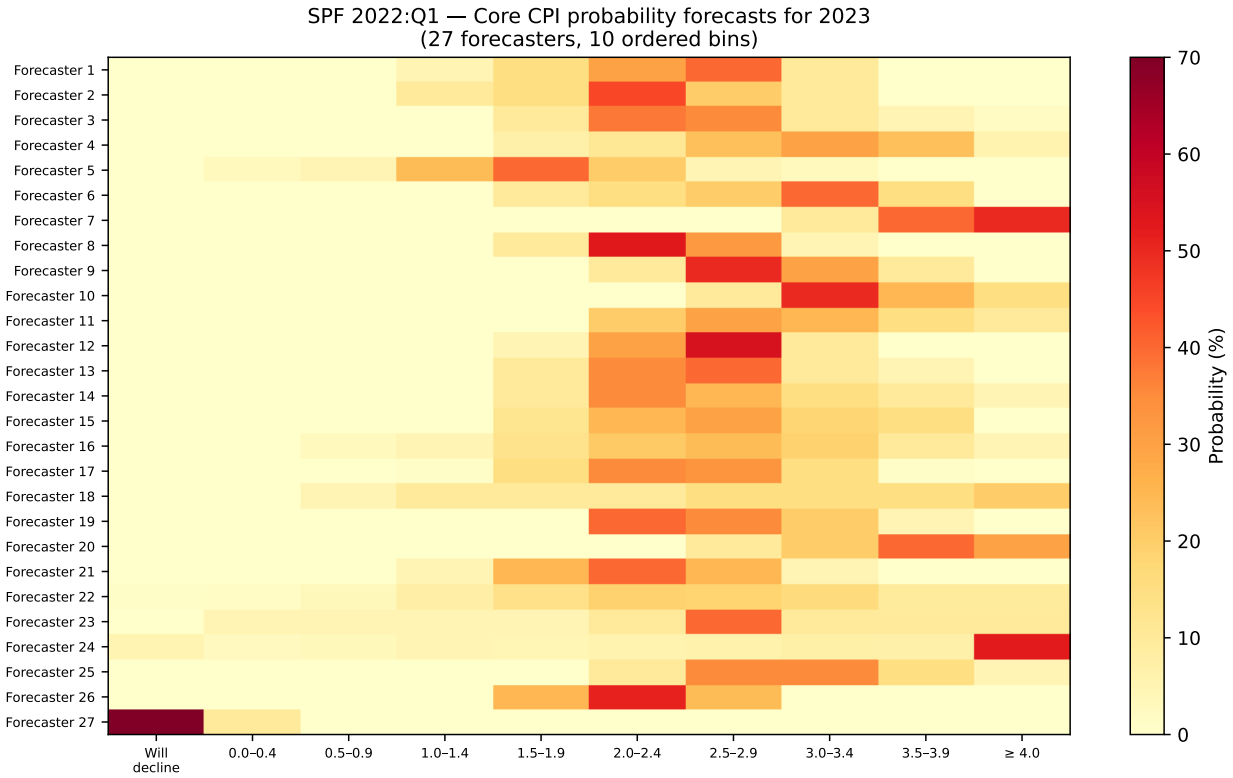

Figure 11: SPF 2022:Q1: probability distributions for 2023 core CPI inflation across 27 professional forecasters. Each row is one forecaster's histogram over 10 ordered bins (ascending from "will decline" to $\geq 4\%$). Cell color encodes probability mass (darker = higher).

ability from the 2–3% bin to the 3–4% bin incurs less transport cost than shifting it to the deflation bin, correctly reflecting the asymmetry in the outcomes. By Proposition 2.3, the W_1 barycenter equals the PMF obtained by component-wise CDF median. We verified this numerically: the closed-form formula and the LP solver agree to machine precision (ℓ^1 discrepancy $< 10^{-4}$).

Figure 12 compares the arithmetic mean and the W_1 barycenter. The differences are substantial. The arithmetic mean places 3.6% mass on the "decline" bin and 8.1% on the $\geq 4\%$ bin, for a combined tail probability of 11.7%. The barycenter assigns *zero* mass to both tails. Instead, it concentrates 60% of its mass on the 2.0–2.9% range and 30% on 3.0–3.9%, reflecting the view shared by a majority of forecasters. The arithmetic mean disperses mass across all bins because a minority of extreme views pull it toward the tails.

Figure 13 shows the individual CDFs (grey), the arithmetic-mean CDF (red), and the

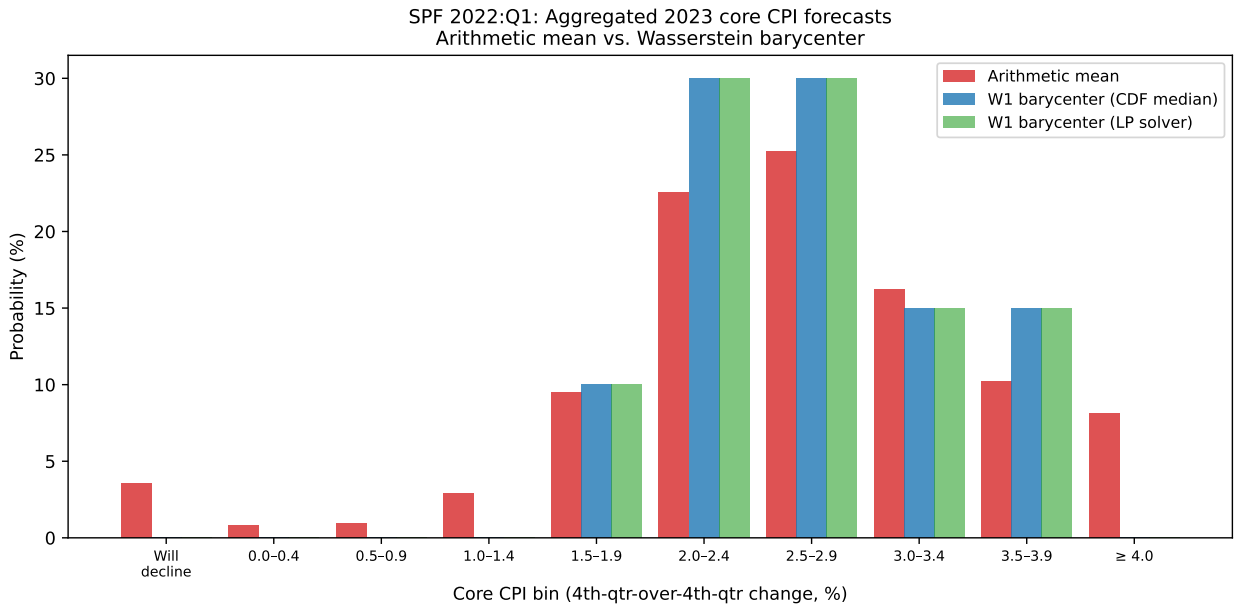

Figure 12: Arithmetic mean (red), W_1 barycenter via CDF-median formula (blue), and W_1 barycenter via LP solver (green) for 2023 core CPI, SPF 2022:Q1. The closed-form and LP results coincide exactly.

barycenter CDF (blue). The barycenter CDF is a step function whose steps track the *vertical median* of the individual CDFs at each bin boundary. The AM CDF is smoother but pulled outward toward the tails by outlying forecasters.

6.3 Historical Forecast Evaluation

Evaluation protocol

Forecast variable. We use the current-year PRCCPI histograms (columns `PRCCPI1`–`PRCCPI10`), in which each respondent distributes 100 probability points across 10 ordered bins for Q4/Q4 core CPI inflation in the current calendar year. The horizon is current-year (not next-year); we choose this horizon because the realized value is unambiguously assigned within the same calendar year and the panel is largest.

Sample construction. The sample covers all survey quarters from 2007:Q1 through 2025:Q4. A respondent's row is retained if its bin probabilities sum to within 2 percentage points of 100; retained histograms are renormalized to exactly 1 and reversed to ascending

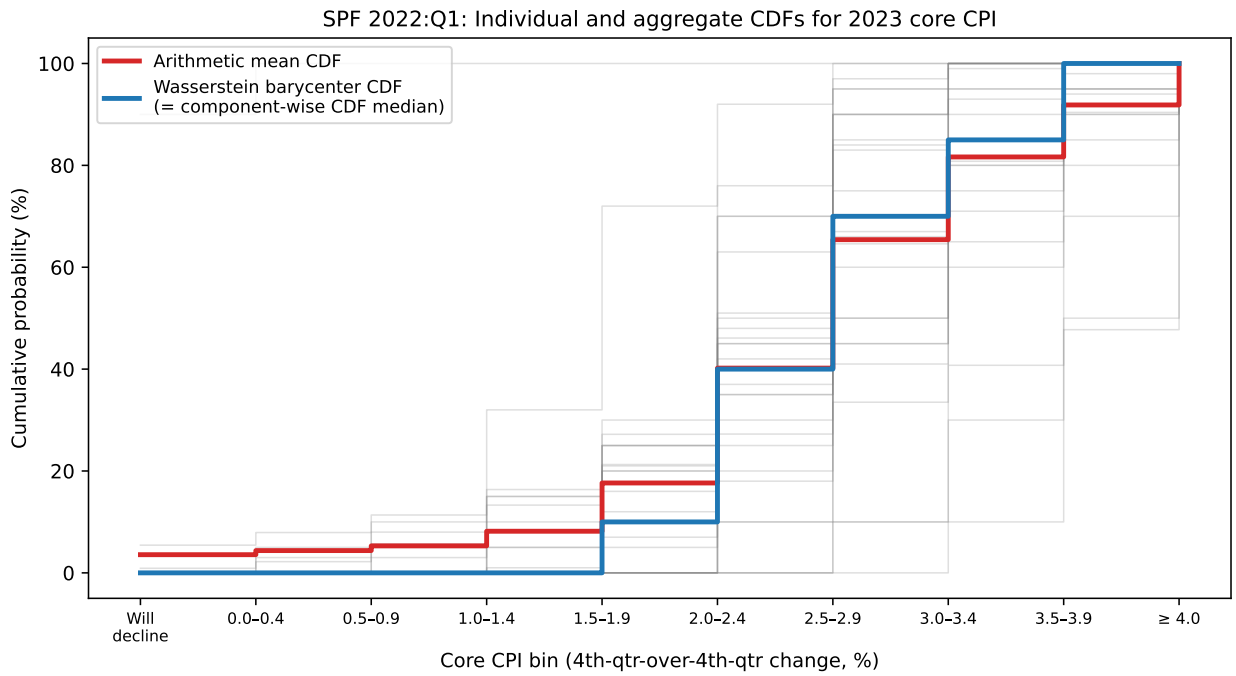

Figure 13: Individual CDFs (grey, $n = 27$) and aggregate CDFs for the arithmetic mean (red) and the W_1 barycenter (blue). The barycenter CDF is the component-wise median of the individual CDFs, confirming Proposition 2.3.

bin order. Quarters with fewer than five valid respondents are excluded, leaving $n = 76$ usable quarters.

Realized outcomes. Each realized value is the December-over-December percent change in seasonally adjusted core CPI (FRED series `CPILFESL`) for the forecast year. The SA and NSA series are nearly identical for this variable. Each realized value is mapped to one of the 10 bins using the boundaries in Table 5.

Scoring rules. The primary scoring rule is the *Ranked Probability Score* (RPS) [13]: RPS $= \sum_{k=1}^{K-1}(F_{\hat{p}}(k) - \mathbf{1}_{b \le k})^2$, where $F_{\hat{p}}$ is the aggregate CDF and b is the realized bin. Lower RPS is better. The secondary criterion is the probability mass the aggregate places on the realized bin (mass score); higher is better. The two criteria are complementary: the mass score rewards precision at the correct bin, while RPS additionally penalizes mass placed far from the realized outcome. A log score is computed for robustness but not the primary criterion because it becomes very large when either method assigns near-zero mass to the realized bin.

Inference. Score differences are tested via one-sided Wilcoxon signed-rank tests, which are robust to non-normality and heavy tails—relevant given the extreme score losses in 2021–2022. We do not evaluate calibration, which is a property of the individual expert forecasts and is not in general preserved under nonlinear combination; a reliability-diagram analysis is left for future work.

We evaluate each method on two complementary scoring criteria. The *Ranked Probability Score* (RPS) [13] is a strictly proper scoring rule for ordered categorical outcomes defined as $\text{RPS} = \sum_{k=1}^{K-1}(F_{\hat{p}}(k) - \mathbf{1}_{b \leq k})^2$, where $F_{\hat{p}}$ is the aggregate CDF and b is the realized bin; lower RPS indicates better accuracy. The *probability on the realized bin* (mass score) is the probability mass the aggregate assigns to the bin containing the realized outcome; higher is better. The two criteria are complementary: the mass score rewards placing mass in precisely the right bin, while RPS additionally penalizes mass placed far from the realized bin. Significance is assessed via one-sided Wilcoxon signed-rank tests, which are robust to non-normality and heavy tails in score differences—a relevant concern given that both methods assign near-zero mass to the $\geq 4\%$ bin in most quarters, causing extreme log-score losses in the 2021–2022 inflation episode.

The single-quarter illustration above motivates a question of practical importance: does the W_1 barycenter produce better probabilistic forecasts than the arithmetic mean when evaluated against realized outcomes?

Overall results. Panel A of Table 5 reports both metrics over the full sample. The most direct finding is that the barycenter places more mass on the correct bin in 75.0% of quarters ($p < 0.001$, Wilcoxon), with a mean probability of 0.424 versus 0.367 for the arithmetic mean, a 15.6% improvement. The RPS comparison tells a consistent story: the barycenter achieves lower RPS in 68.4% of quarters ($p = 0.0045$), with mean RPS 0.439 versus 0.447.

Dispersion-conditional results. Panel B conditions on the cross-sectional variance of individual forecast means, splitting the sample at the median into low- and high-dispersion

quarters (38 each). The probability-on-realized-bin metric shows the barycenter outperforming in *both* groups: 0.496 versus 0.430 in low-dispersion quarters, and 0.353 versus 0.303 in high-dispersion quarters. The RPS results are more nuanced: the barycenter wins decisively in low-dispersion quarters (mean RPS 0.301 vs. 0.328, $p < 0.001$), while the arithmetic mean is marginally better in high-dispersion quarters (mean RPS 0.567 vs. 0.578, $p = 0.37$). The divergence between the two metrics in high-dispersion quarters reflects a distinction in what they reward. The barycenter places more mass on the realized bin even in these quarters, but when the realized bin is a tail bin, the AM's broader CDF has smaller cumulative deviations from the realized step function, yielding lower RPS. The 2021–2022 inflation surge is the clearest illustration: both methods underestimated realized core CPI ($\geq 4\%$), but the AM retained more tail mass, reducing its RPS deficit in those extreme quarters.

In low-dispersion quarters the economic interpretation is straightforward: the panel agrees, the barycenter correctly concentrates mass on the consensus region, and the consensus is usually right. In high-dispersion quarters the panel is genuinely uncertain; the arithmetic mean's equal weighting preserves more tail mass, which pays off under RPS when an extreme outcome materializes, even though the barycenter still places more mass on the correct bin in expectation.

Table 5: SPF current-year backtest: arithmetic mean (AM) vs. W_1 barycenter (BC), 2007:Q1–2025:Q4. "Mass on correct bin": probability assigned to the realized bin (higher is better). RPS: lower is better. p-values from one-sided Wilcoxon signed-rank test.

	Mass on correct bin		Mean RPS	
Sample	AM	BC	AM	BC
Panel A: Full sample ($n = 76$ quarters)				
All quarters	0.367	0.424	0.447	0.439
BC better	75.0% of quarters		68.4% of quarters	
p-value	<0.001		0.0045	
Panel B: By cross-sectional dispersion ($n = 38$ each)				
Low dispersion	0.430	0.496	0.328	0.301
High dispersion	0.303	0.353	0.567	0.578

Figures 14–17 provide the visual summary. Figure 14 shows the probability mass assigned

to the correct bin in each quarter; the barycenter's advantage is visible throughout the sample and in both dispersion groups. Figure 15 shows the RPS time series for both methods. Figure 16 plots the quarter-by-quarter RPS difference, with the high-dispersion quarters (shaded) containing the episodes where the arithmetic mean recovers ground. Figure 17 summarizes the dispersion-conditional RPS comparison.

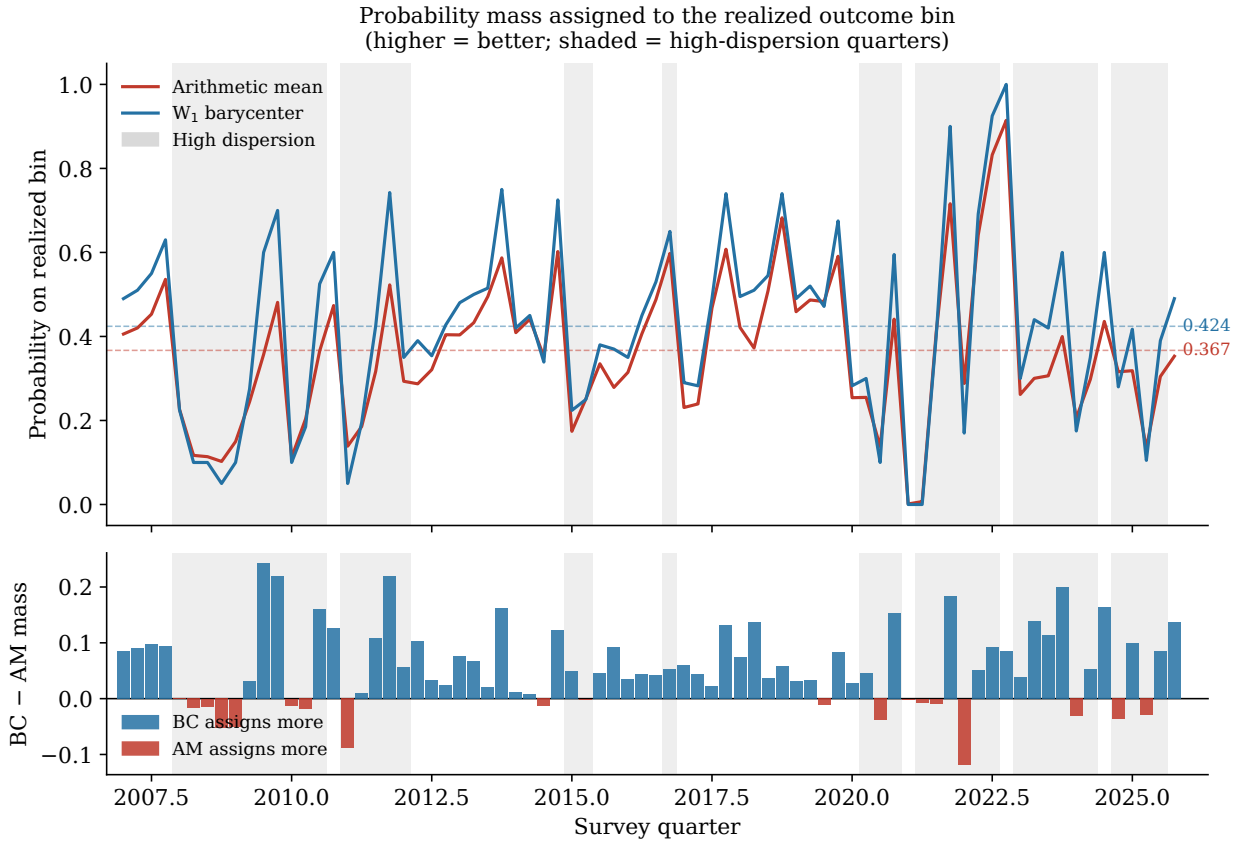

Figure 14: Top: probability mass assigned to the realized outcome bin by the arithmetic mean (red) and W_1 barycenter (blue), 2007:Q1–2025:Q4. Dashed lines mark the sample means (AM: 0.367; BC: 0.424). Shaded regions are high-dispersion quarters. Bottom: quarter-by-quarter difference (BC − AM); positive bars (blue) indicate quarters where the barycenter places more mass on the correct bin.

6.4 Adaptive Blending

The historical results suggest a natural extension. The W_1 barycenter improves on the arithmetic mean on average, but the arithmetic mean retains value in high-dispersion quarters

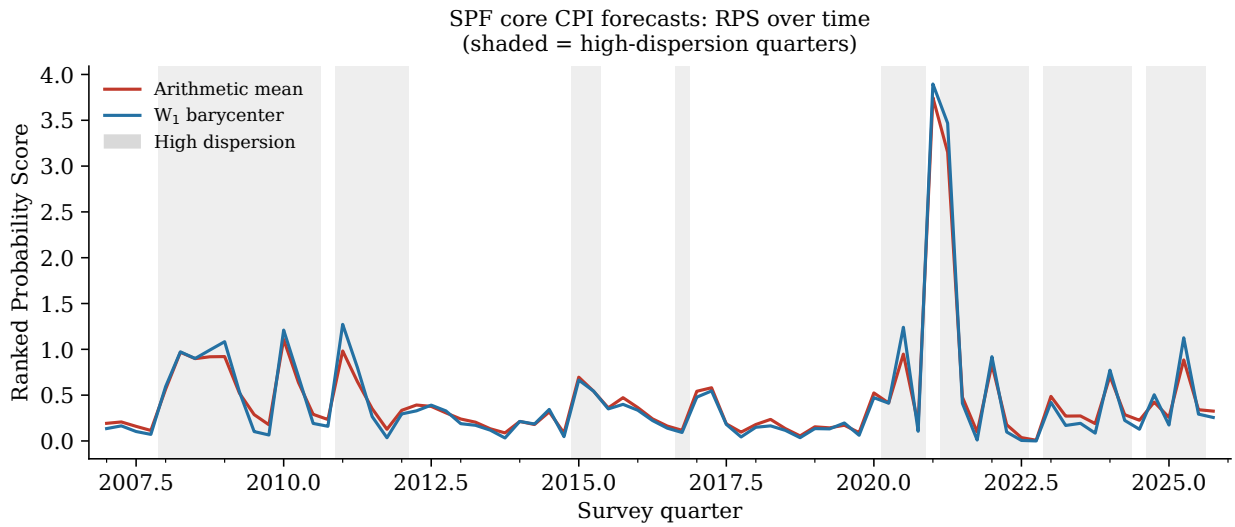

Figure 15: AM (red) and W_1 barycenter (blue) RPS over 76 survey quarters, 2007:Q1–2025:Q4. Shaded regions mark high-dispersion quarters (cross-sectional variance of forecast means above the sample median).

under the RPS. A convex blend

$$\hat{\Psi}_t(\lambda) \;=\; \lambda\,\Psi_t^* + (1-\lambda)\,\bar{P}_t, \qquad \lambda \in [0,1],$$

interpolates between the two extremes. At $\lambda = 1$ this is the barycenter; at $\lambda = 0$ it is the arithmetic mean.

Oracle. A grid search over $\lambda \in \{0, 0.005, \ldots, 1\}$ using all 76 quarters identifies $\lambda^* = 0.70$ as the mean-RPS minimizer, yielding an in-sample mean RPS of 0.437—lower than both the arithmetic mean (0.447) and the pure barycenter (0.439). The improvement is modest but reveals that neither extreme is optimal.

Out-of-sample evaluation. To avoid look-ahead bias we apply an expanding-window protocol: for each quarter t (starting from quarter 21) we estimate $\hat{\lambda}_t$ by minimizing mean RPS over all prior quarters and apply it only to quarter t. In a richer adaptive version, $\hat{\lambda}_t = \sigma(a_t + b_t D_t)$ where D_t is the cross-sectional variance of individual forecast means and (a_t, b_t) are estimated by Nelder-Mead on the expanding window. Results over the 56-quarter

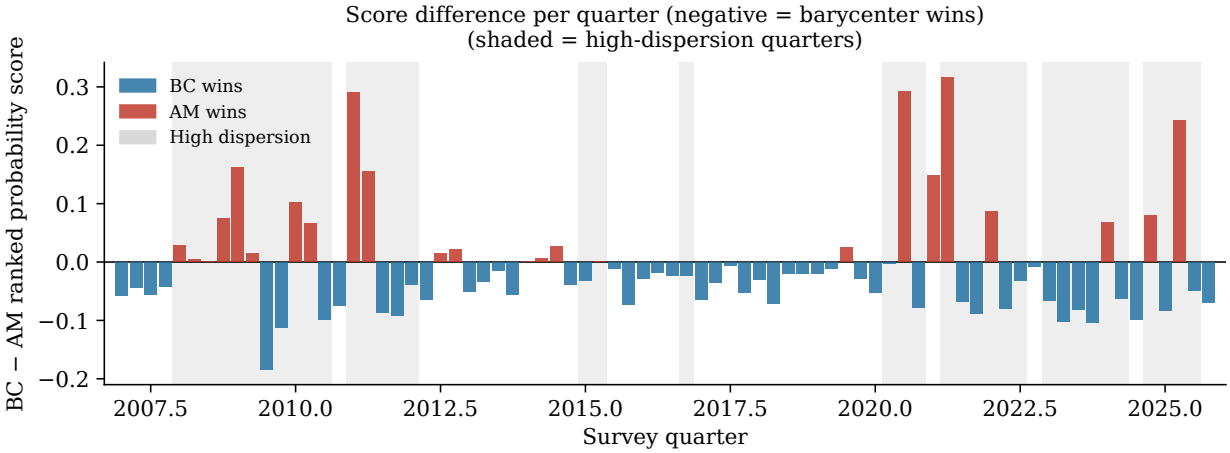

Figure 16: Quarter-by-quarter RPS difference (BC − AM). Negative bars (blue) indicate quarters where the barycenter achieves a lower (better) RPS; positive bars (red) indicate quarters where the arithmetic mean wins. The barycenter outperforms in 68.4% of quarters.

out-of-sample period are reported in Table 6 and Figure 18.

Method	Mean RPS	p-value (vs. AM)	Wins vs. AM
Arithmetic mean	0.423	—	—
W_1 barycenter	0.411	0.002	75%
Fixed blend ($\hat{\lambda} \approx 0.71$)	0.413	0.001	79%
Adaptive blend	0.404	<0.001	64%

Table 6: Out-of-sample RPS comparison (56 quarters, expanding-window evaluation). p-values from Wilcoxon signed-rank test. The dispersion-adaptive blend achieves the lowest mean RPS.

The adaptive blend achieves a mean RPS of 0.404—a 4.5% improvement over the arithmetic mean and a 1.7% improvement over the unweighted barycenter. The expanding-window estimate of λ converges rapidly to approximately 0.71, closely matching the oracle value, which indicates that the optimal mixture weight is stable over time and requires no large calibration sample.

6.5 Replication on the ECB Survey of Professional Forecasters

To assess whether the US SPF findings are specific to that survey or reflect a more general property of barycenter aggregation, we replicate the backtest on the ECB Survey of Profes-

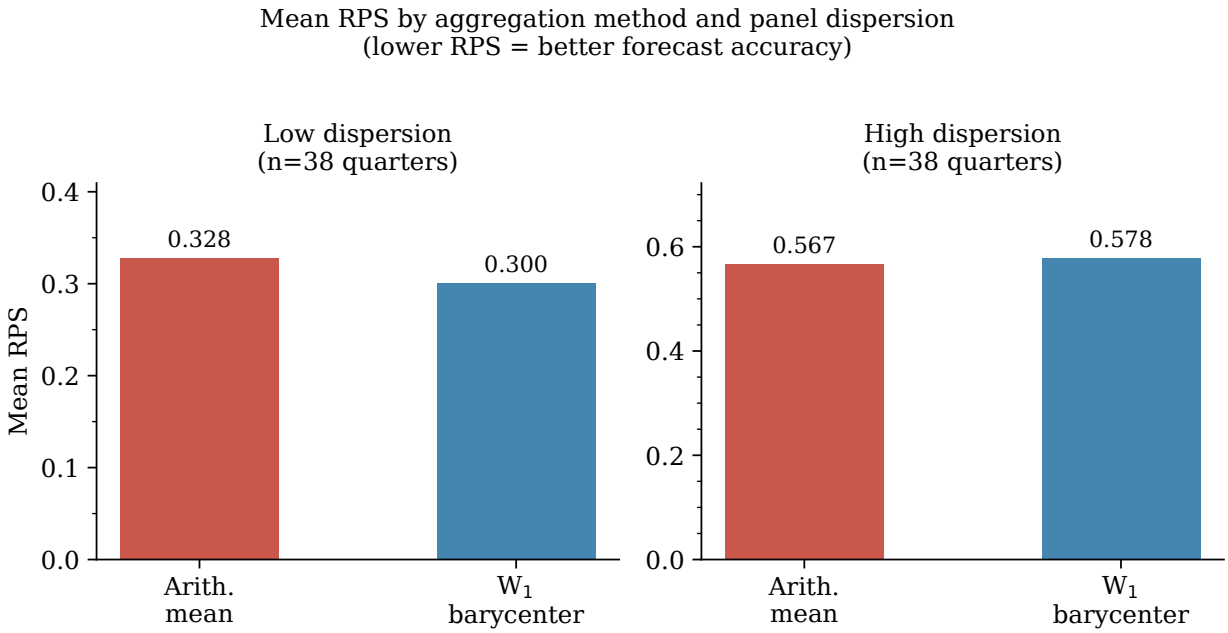

Figure 17: Mean RPS by aggregation method and dispersion group. In low-dispersion quarters the barycenter reduces mean RPS by 8.3% relative to the arithmetic mean; in high-dispersion quarters the arithmetic mean is marginally better.

sional Forecasters (ECB SPF), which has been conducted quarterly since 1999 [1]. The ECB SPF covers the Euro area and asks respondents to assign probabilities to ordered bins for headline HICP inflation, real GDP growth, and the unemployment rate. We use the headline HICP current-year forecasts over the period 2010Q1–2024Q3 ($n = 56$ quarter-surveys), during which the bin structure is stable at 12 ordered bins of width 0.5 pp from below -1% to above 4%. Realized values are annual-average HICP YoY from the ECB Statistical Data Warehouse. The evaluation protocol (AM, W_1 barycenter, RPS scoring, dispersion split) is identical to Section 6.3.

Results. The barycenter assigns more probability mass to the realized outcome bin in 78.6% of quarter-surveys ($p < 0.001$, Wilcoxon), with a mean mass of 0.457 versus 0.430 for the arithmetic mean—a 6.3% improvement (Figure 19). This replicates the core finding from the US SPF on an independent panel in a different institutional setting and with a different realized variable.

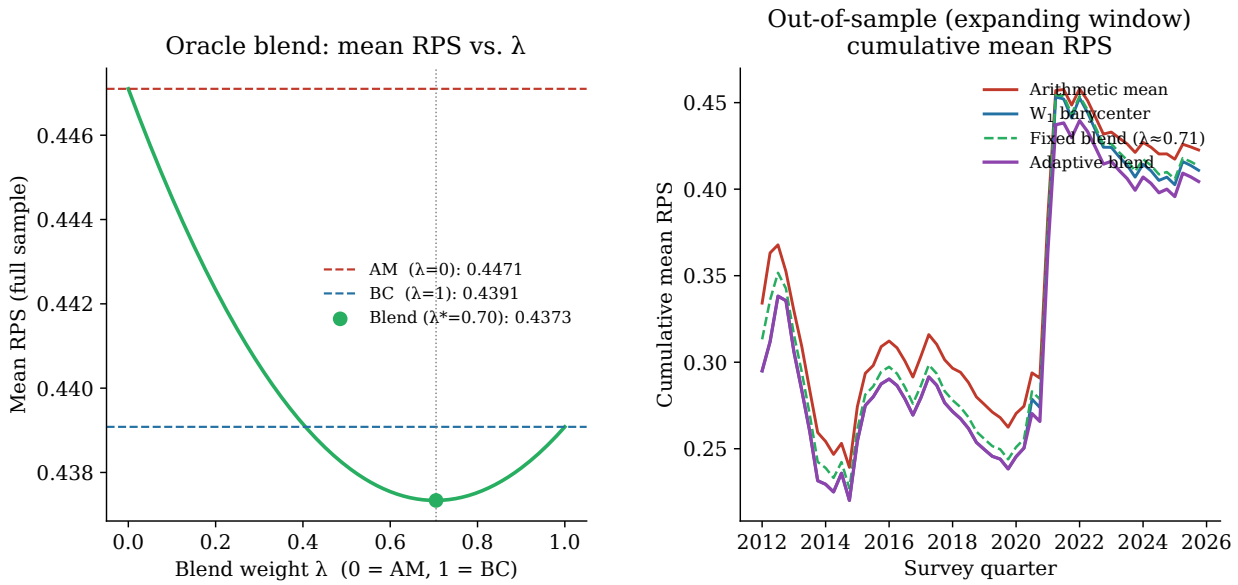

Figure 18: Left: oracle mean RPS as a function of blend weight λ (full sample). The minimum at $\lambda^* = 0.70$ lies between the arithmetic mean ($\lambda = 0$) and the barycenter ($\lambda = 1$). Right: cumulative out-of-sample mean RPS for the arithmetic mean, W_1 barycenter, fixed blend, and dispersion-adaptive blend over the 56-quarter evaluation period.

The aggregate RPS advantage, however, is small and not statistically significant (mean RPS 0.441 AM vs. 0.441 BC, $p = 0.23$). The dispersion-conditional pattern is consistent with the US SPF: the barycenter achieves lower RPS in low-dispersion quarters (mean 0.255 vs. 0.266) while the arithmetic mean wins marginally in high-dispersion quarters (mean 0.616 vs. 0.626). The divergence between the mass-on-correct-bin and RPS metrics is explained by the 2021–2023 Euro-area inflation surge. Annual-average HICP reached 8.4% in 2022—well above the highest bin boundary ($\geq$4%)—an outcome that both methods failed to anticipate but that the arithmetic mean handles less badly under RPS by retaining more probability mass in the upper tail of the distribution.

Interpretation. The ECB SPF replication confirms that the barycenter's concentration near the panel consensus is a reliable property, not a sample artifact: it places more mass on the realized bin in both the US and Euro-area panels. The non-significant RPS result for the ECB sample reflects the particular severity of the 2022 inflation episode, whose outcome

	Mean score		
Method	RPS	P(correct bin)	Wins (mass)
Arithmetic mean	0.441	0.430	—
W_1 barycenter	0.441	0.457	78.6%***
Panel B: By dispersion ($n = 28$ each)			
	Mean RPS		
	Low disp.	High disp.	
Arithmetic mean	0.266	0.616	
W_1 barycenter	0.255	0.626	

Table 7: ECB SPF backtest (56 quarter-surveys, 2010Q1–2024Q3, current-year headline HICP). *** Wilcoxon $p < 0.001$. Overall RPS difference is not significant ($p = 0.23$).

lay far outside the normal support of the distribution. Taken together, the two backtests suggest that the barycenter gains from sharper concentration near the consensus and can lose from reduced tail mass in extreme-realization quarters.

7 Discussion

This paper proposes the Wasserstein barycenter as a robust method for aggregating expert probability forecasts over finite outcome spaces. The central theoretical result is Theorem 2.2: under the indicator ground cost, the barycenter reduces to a constrained ℓ^1 aggregation problem. This simplification has two important consequences. First, it makes computation straightforward: the problem is a convex linear program in NK variables, solvable by standard LP solvers; for ordered spaces, Proposition 2.3 gives a fully closed-form answer (component-wise CDF median) with no solver required. Second, it identifies the barycenter as a median-type aggregate: the unconstrained minimizer of the ℓ^1 objective assigns the coordinatewise median to each state, and the simplex constraint shifts this toward a normalized version of the median. For ordered outcome spaces the closed form is exact: the W_1 barycenter is the distribution whose CDF is the component-wise median of the individual CDFs (Proposition 2.3), a result verified numerically to machine precision in the SPF application.

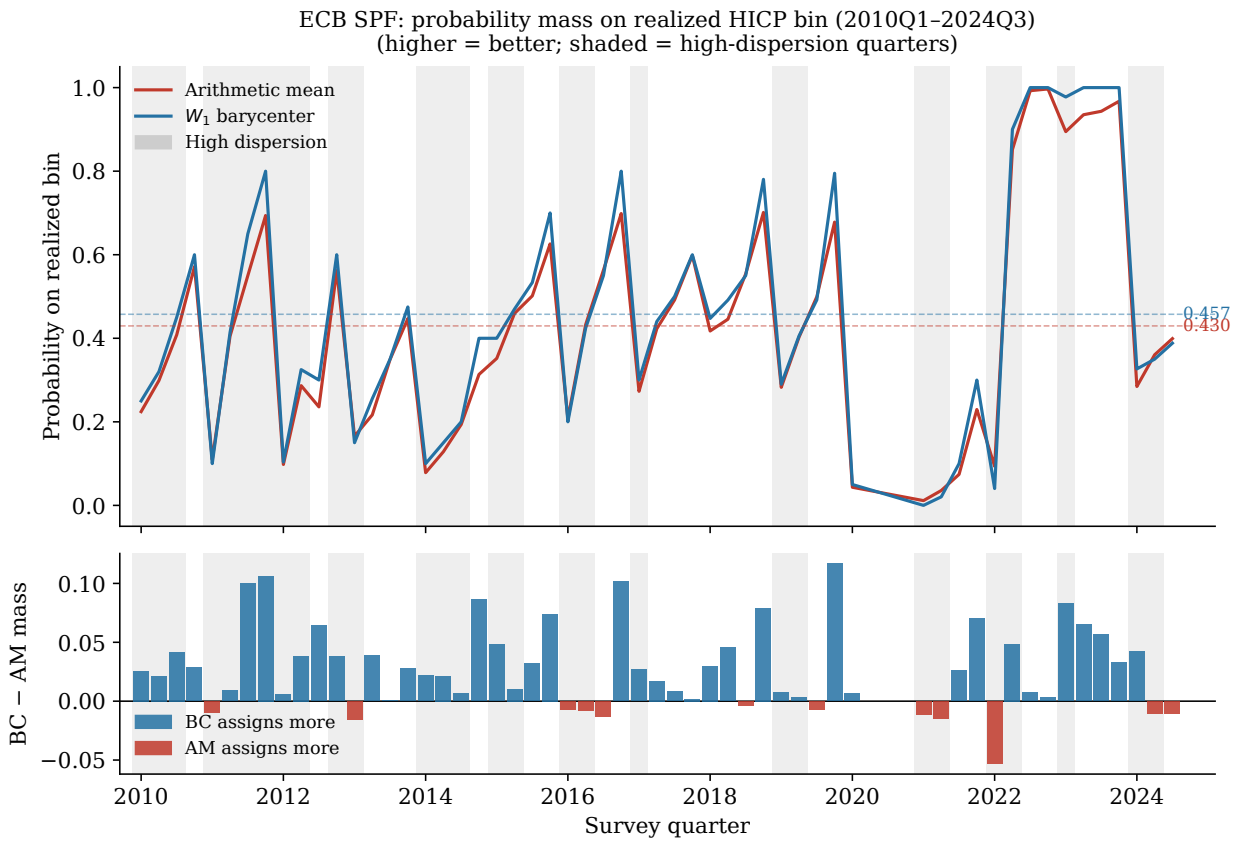

Figure 19: ECB SPF: probability mass assigned to the realized annual-average HICP bin by the arithmetic mean and W_1 barycenter, 2010Q1–2024Q3. The barycenter consistently assigns more mass to the correct bin. Shaded quarters are high-dispersion (above-median cross-sectional variance). The 2022 inflation surge (realized HICP 8.4%) is visible as a period of low mass for both methods.

Robustness interpretation. The EIF analysis formalizes the intuition that the barycenter is less sensitive than the arithmetic mean to extreme opinions. The AM gives every expert equal weight regardless of distance from the consensus; the barycenter penalizes outlying opinions through transport cost, so an expert far from the current aggregate exerts smaller influence. Empirically, in 90% of simulation replicates the barycenter assigns less probability to an outlier's preferred state than the AM. The tradeoff is efficiency: under a well-specified homogeneous expert population, the AM has lower MSE by a factor of approximately 1.4–1.9.

Forecast evaluation. The historical backtest in Section 6.3 provides direct evidence that the theoretical properties of the barycenter translate into empirical gains. The barycenter assigns more probability mass to the realized outcome bin in a majority of quarters, with a mean improvement of 15.6% (0.424 vs. 0.367), and achieves a lower average Ranked Probability Score than the arithmetic mean. The advantage is concentrated in low-dispersion quarters; the arithmetic mean recovers ground under RPS in high-dispersion quarters because its broader tail coverage reduces cumulative CDF deviations when extreme outcomes materialize. A dispersion-adaptive convex blend of the two methods achieves the lowest out-of-sample RPS (0.404), versus 0.411 for the pure barycenter and 0.423 for the arithmetic mean. The optimal mixture weight ($\lambda^* \approx 0.70$) is stable over time, matching the oracle value within the expanding-window evaluation. These results together characterize when each method is preferred: the barycenter is the better aggregator when the panel is broadly in agreement, while the arithmetic mean's equal weighting preserves tail mass that is occasionally valuable when extreme outcomes materialize. The Wasserstein barycenter is not calibrated by construction—calibration is a property of the individual expert forecasts and is not in general preserved under nonlinear combination; a full calibration analysis (e.g., PIT histograms or reliability diagrams) is left for future work. Comparing the barycenter to alternative methods such as the logarithmic pool and extremization [17], and extending the evaluation to multi-horizon forecasts and other outcome variables, are natural further directions.

Applications and extensions. The two applications illustrate the method on both unordered (opioid survey, indicator distance) and ordered (SPF, W_1 distance) outcome spaces. A natural extension is to weighted barycenters, in which experts receive unequal weights reflecting their track record or domain expertise. A natural population model for expert probability vectors is the Dirichlet distribution; however, aggregation via Dirichlet MLE does not generally satisfy boundedness or unanimity, and therefore differs conceptually from

the deterministic aggregation rules studied here. The extension to probability distributions over network-valued outcomes (Appendix A) illustrates that the framework applies to any finite metric space without modification. The multivariate case, in which experts provide joint distributions over multiple outcomes, raises new questions about the structure of the barycenter that we leave for future work.

A Barycenters for Network-Valued Outcomes

The framework extends naturally to settings where the outcome space is a set of network structures. Let $\mathcal{Y}_n$ denote the space of binary adjacency matrices on n nodes: the set of $n \times n$ zero-diagonal matrices with entries in $\{0, 1\}$, equipped with the Hamming distance

$$d(Y, Y') = \sum_{m \neq n} |Y_{mn} - Y'_{mn}| \,.$$

A probability mass function over $\mathcal{Y}_n$ assigns probability $p_j(Y)$ to each network $Y \in \mathcal{Y}_n$. Existence of the barycenter follows from the compactness argument of Theorem 2.2 applied to the finite metric space $(\mathcal{Y}_n, d)$. The full-diagonal criterion (Proposition B.1) carries over: in any optimal solution, $\Pi^{(j)}_{YY} = \min\{p_j(Y), \Psi(Y)\}$ for each Y.

Unlike the unordered case where all off-diagonal pairs are equidistant, pairs of networks differing in k edges incur cost k under W_1 (or k^2 under W_2). After diagonal entries are maximized, remaining mass is assigned to off-diagonal entries in order of increasing Hamming distance.

The result extends to integer-valued (weighted) networks by replacing $\{0, 1\}$ with $\{0, \ldots, M\}$ and using the ℓ^1 Hamming distance.

B Full-Diagonal Criterion

The following proposition extends Lemma 2.1 from individual transport plans to the full barycenter solution: in any optimal solution, every transport plan simultaneously maximises its diagonal.

Proposition B.1 (Full-Diagonal Criterion)**.** *Let $\Pi^1, \ldots, \Pi^N$ be $K \times K$ matrices representing the joint distributions between a candidate barycenter Ψ and the probability mass functions $p_1, \ldots, p_N$ on $\Omega = \{1, \ldots, K\}$. Suppose the matrices are feasible: for all l and k,*

$$\sum_j \Pi^l_{k,j} = p_l(k), \qquad \sum_j \Pi^l_{j,k} = \Psi(k),$$

but for some l and i,

$$\Pi^l_{i,i} < \min\big\{p_l(i),\, \Psi(i)\big\}. \tag{B.1}$$

Then there exists a feasible $\tilde{\Pi}^l \neq \Pi^l$ such that $f(\tilde{\Pi}^l, \Pi^{-l}) < f(\Pi^l, \Pi^{-l})$, so $(\Pi^1, \ldots, \Pi^N)$ is not optimal. Consequently, every optimal transport plan satisfies $\Pi^l_{kk} = \min\{p_l(k), \Psi(k)\}$ for all k.

Proof. The objective equals $f(\mathbf{\Pi}) = \sum_{l=1}^N \sum_{k=1}^K (1 - \Pi^l_{k,k})$, so increasing any diagonal entry strictly decreases the objective. Suppose (B.1) holds. Since $\Pi^l_{ii} < p_l(i)$, there exists $k_1 \neq i$ with $\Pi^l_{i,k_1} > 0$. Setting $\tilde{\Pi}^l_{i,i} = \Pi^l_{i,i} + \Pi^l_{i,k_1}$ and $\tilde{\Pi}^l_{i,k_1} = 0$ preserves row i's sum. Since $\Pi^l_{ii} < \Psi(i)$, the column-i sum $\Psi(i)$ exceeds the diagonal; there exists a set of indices $\{k_2^i\}$ and weights $\{\epsilon_i\}$ summing to Π^l_{i,k_1} such that transferring mass $\epsilon_i \Pi^l_{k_2^i,i}$ from column i to column k_1 restores all column sums. The resulting $\tilde{\Pi}^l$ is feasible and the objective decreases by $\Pi^l_{i,k_1} > 0$. $\square$

References

[1] Juan Angel Garcia and André Manzanares. *What Can Probability Forecasts Tell Us About the Risks to Output and Inflation?* European Central Bank Working Paper No. 483, 2005.

[2] Philip E. Tetlock and Dan Gardner. *Superforecasting: The Art and Science of Prediction*. Crown Publishers, New York, 2016.

[3] Philip E. Tetlock, Barbara A. Mellers, Nick Rohrbaugh, and Eva Chen. *Good Judgment Project: Four Years of Forecasts*. Harvard Dataverse, V3, doi:10.7910/DVN/BPCDH5, 2014.

[4] Robert T. Clemen. *Combining Forecasts: A Review and Annotated Bibliography*. International Journal of Forecasting, 5(4):559–583, 1989.

[5] Roopesh Ranjan and Tilmann Gneiting. *Combining Probability Forecasts*. Journal of the Royal Statistical Society, Series B, 72(1):71–91, 2010.

[6] Tilmann Gneiting and Roopesh Ranjan. *Combining Predictive Distributions*. Electronic Journal of Statistics, 7:1747–1782, 2013.

[7] Kajal Lahiri, Herman Peng, and Xuguang Sheng. *Measuring Uncertainty of a Combined Forecast and Some Tests for Forecaster Heterogeneity*. International Journal of Forecasting, 31(1):113–129, 2015.

[8] Xiaoqian Wang, Rob J. Hyndman, Feng Li, and Yanfei Kang. *Forecast Combinations: An Over 50-Year Review*. International Journal of Forecasting, 39(4):1518–1547, 2023.

[9] Thomas McAndrew, Nutcha Wattanachit, Graham C. Gibson, and Nicholas G. Reich. *Aggregating Predictions from Experts: A Review of Statistical Methods, Experiments, and Applications*. WIREs Computational Statistics, 13(2):e1514, 2021.

[10] Camille Le Coz, Alexis Tantet, Rémi Flamary, and Riwal Plougonven. *A Barycenter-Based Approach for the Multi-Model Ensembling of Subseasonal Forecasts*. arXiv:2310.17933, 2023.

[11] James W. Taylor. *Probabilistic Forecast Aggregation with Statistical Depth*. European Journal of Operational Research, 328(2):460–476, 2026.

[12] C. Genest and J.V. Zidek. *Combining Probability Distributions: A Critique and an Annotated Bibliography*. Statistical Science, 1(1):114–135, 1986.

[13] Tilmann Gneiting and Adrian E. Raftery. *Strictly Proper Scoring Rules, Prediction, and Estimation*. Journal of the American Statistical Association, 102(477):359–378, 2007.

[14] A. O'Hagan, C.E. Buck, A. Daneshkhah, J.R. Eiser, P.H. Garthwaite, D.J. Jenkinson, J.E. Oakley, and T. Rakow. *Uncertain Judgements: Eliciting Experts' Probabilities*. John Wiley & Sons, Chichester, 2006.

[15] S.S. Vallender. *Calculation of the Wasserstein Distance Between Probability Distributions on the Line*. Theory of Probability and Its Applications, 18(4):784–786, 1973.

[16] Victor M. Panaretos and Yoav Zemel. *Statistical Aspects of Wasserstein Distances*. Annual Review of Statistics and Its Application, 6:405–431, 2019.

[17] Ville A. Satopää, Jonathan Baron, Dean P. Foster, Barbara A. Mellers, Philip E. Tetlock, and Lyle H. Ungar. *Combining Multiple Probability Predictions Using a Simple Logit Model*. International Journal of Forecasting, 30(2):344–356, 2014.

[18] Allan Timmermann. *Forecast Combinations*. In G. Elliott, C.W.J. Granger, and A. Timmermann (Eds.), *Handbook of Economic Forecasting*, Volume 1, pages 135–196. Elsevier, Amsterdam, 2006.

[19] Martial Agueh and Guillaume Carlier. *Barycenters in the Wasserstein Space*. SIAM Journal on Mathematical Analysis, 43(2):904–924, 2011.

[20] Ethan Anderes, Steffen Borgwardt, and Jacob Miller. *Discrete Wasserstein Barycenters: Optimal Transport for Discrete Data*. Mathematical Methods of Operations Research, 84(2):389–409, 2016.

[21] Marco Cuturi and Arnaud Doucet. *Fast Computation of Wasserstein Barycenters*. Proceedings of the 31st International Conference on Machine Learning, Beijing, China, 2014. JMLR: W&CP, Volume 32.